Commissioning RIO EMU for use with Ignition UDTs (OPC UA Version)

Document version 2.0 for use with Ignition UDTs version 2:

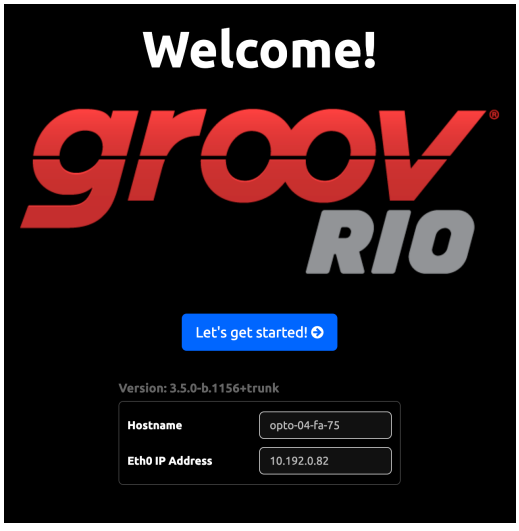
- Folder name: Energy Monitoring UDTs
  - RIO-EMU-OPC_v2

Prerequisites

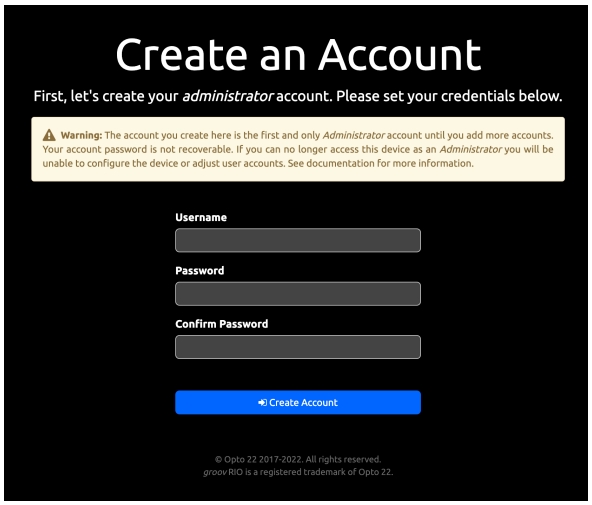
- *groov* RIO EMU - part # GRV-R7-I1VAPM-3
- *groov* EPIC - part # GRV-PR2-EPIC
- Ignition Edge license - part # GROOV-LIC-EDGE8
- Ignition Cloud Edition or on-premises Ignition Standard Gateway (pre-configured and deployed)

Install and Configure the RIO EMU

1. Install RIO EMU and connect to load. For this example, the RIO EMU is not connected to a real load.
   a. Note installed CT load-side ratings (e.g., 1000A, for later configuration in *groov* Manage and Ignition)
   b. Ensure all CT's are facing the same way and are matched to their load phases
   c. Note the wiring method (Delta vs. Wye, for later configuration in *groov* Manage and Ignition)
   d. Connect to a DHCP-enabled network.
   These instructions assume DHCP services are available. For static IP configuration, please see RIO EMU User's Guide.
   e. Connect to power, either line power or PoE
2. Commission RIO EMU user and networking
   a. To connect to the RIO EMU, open your browser (Chrome is used in these instructions)
   b. Enter https://[**default hostname**] (e.g., https://opto-04-fa-75)
   c. Accept security warning
   d. Click the blue button labeled "Let's get started!"



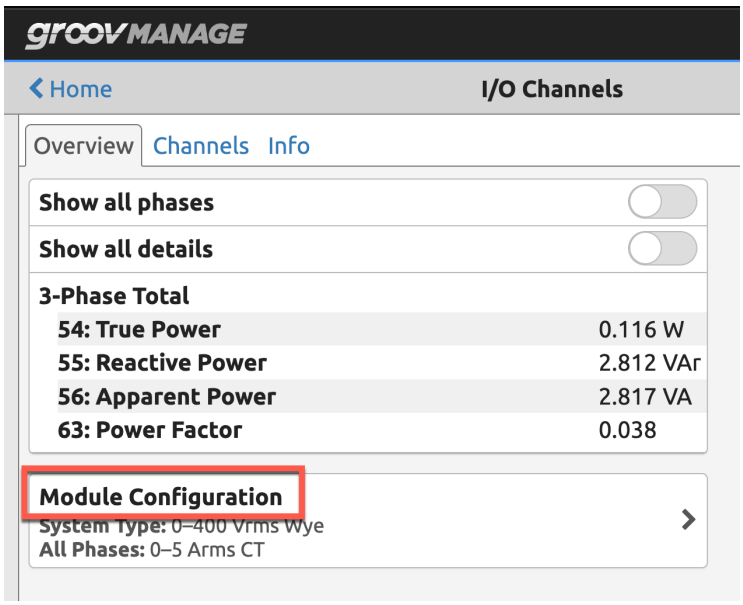
   e. Create an administrator account; note your Username and Password. These cannot be retrieved later.



f. Click the “Configure Device” button
g. Click the blue “System” button
    i. Click the “Time” button
    ii. Click Universal under Time Zone
    iii. Set your timezone Region and Locality
    iv. Click “Set Zone” to apply changes
    v. Confirm your Time settings are now correct
h. Click “System” at the top left, and then “Home” to return to the main menu
i. Click the “Network” button, then the “Configure” button
    i. Click the default hostname value (e.g., opto-04-fa-75) and change it to a network hostname suitable for this load (e.g., MyEMU). All hostnames for multiple EMUs must be unique.
    ii. Click “Save” at the top right
        1. Network services will restart, and your browser will navigate to the new URL with your chosen hostname
j. Accept security warning
k. Enter the just created administrator Username and Password
l. Click the “Home” button to return to the Home page

3. Configure Security Certificates
    a. From the Home page, click the “Security” button
    b. Click the “Web Server Certificate” button
    c. Click the “Create Certificate” button
        i. Complete the resulting form, then click “Create” at the top right corner
            1. Wait for the Certificate to be created, and click “Reload”
        ii. Log back into the RIO EMU
        iii. Click the “Download Public Certificate” and “Download Private Key” buttons to download the Public Certificate and Private Key to a safe place on your PC or network drive
        iv. Upload the Public Certificate to your computer’s Trust Store (see this link for instructions) Note: If you wish to have your certificate signed by a Certificate Authority, click the “Download CSR” button. Send the resulting file to your CA for signing, and upload the signed certificate.
        v. Go back to the home page
4. Configure I/O Channels
    a. From the Home page, click the “I/O” button

b. Click “Module Configuration” near the middle of the page

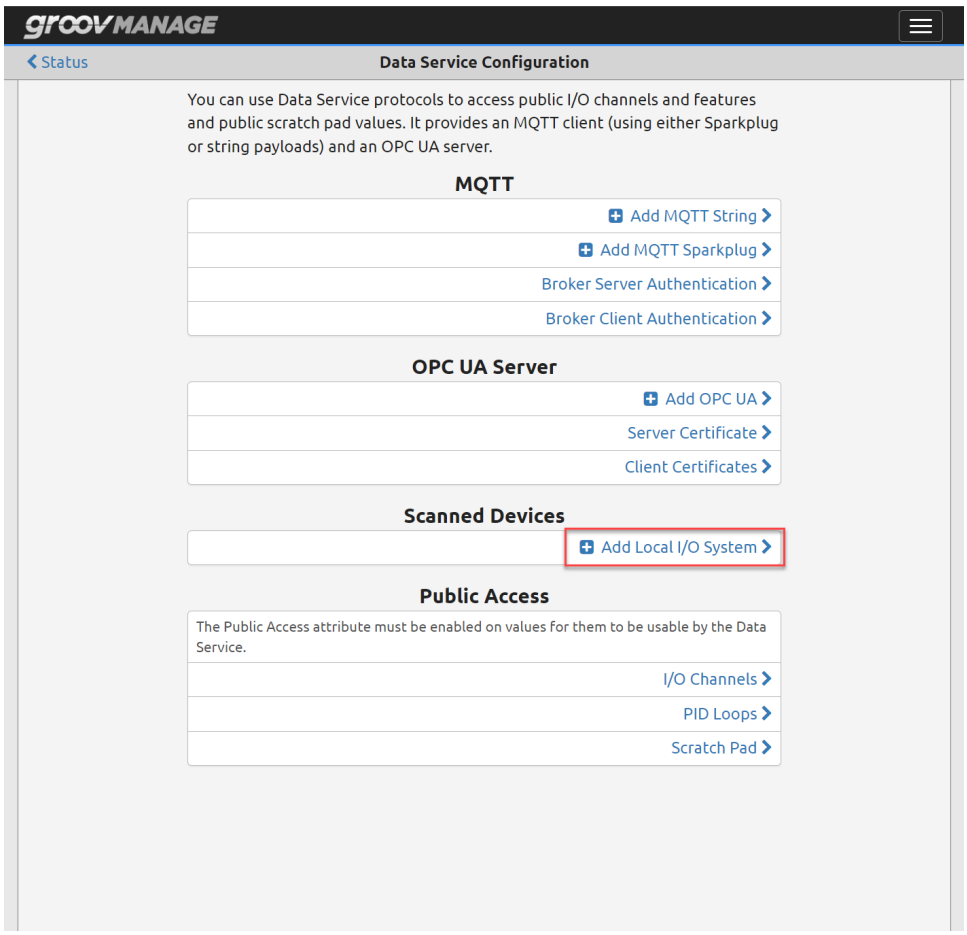


c. Configure System Type by clicking on “0-400 Vrms Wye” and choosing your load’s voltage range and wiring method noted earlier (Delta or Wye)
d. Under Current Transformer, choose your CT Signal Output Type from available options for each phase being used
e. Under CT Rated Current, enter your CT load-side rating noted above (e.g. “1000” for 1000A CTs). If all CT Rated Currents are identical for each phase (most common), you need only enter the value for Phase A.
f. If you’re using a Potential Transformer (for measuring voltages over 600V), enter a turns ratio for the PT in Phase A (all voltage measurements should be the same).
g. Click “Save” in the upper right corner
h. Confirm and validate readings from load under 3-Phase Total. You may also expand Show all phases and Show all details if you wish.
i. Click the “I/O Services” button near the bottom of the page
j. Click the “I/O Batch Operations” button
k. The Operation should read “Configure Public Access”
    i. Ensure the Options reading “State/Value (Read)” slider is ON.
    ii. Click the “Execute” button
l. Navigate back to the Home page

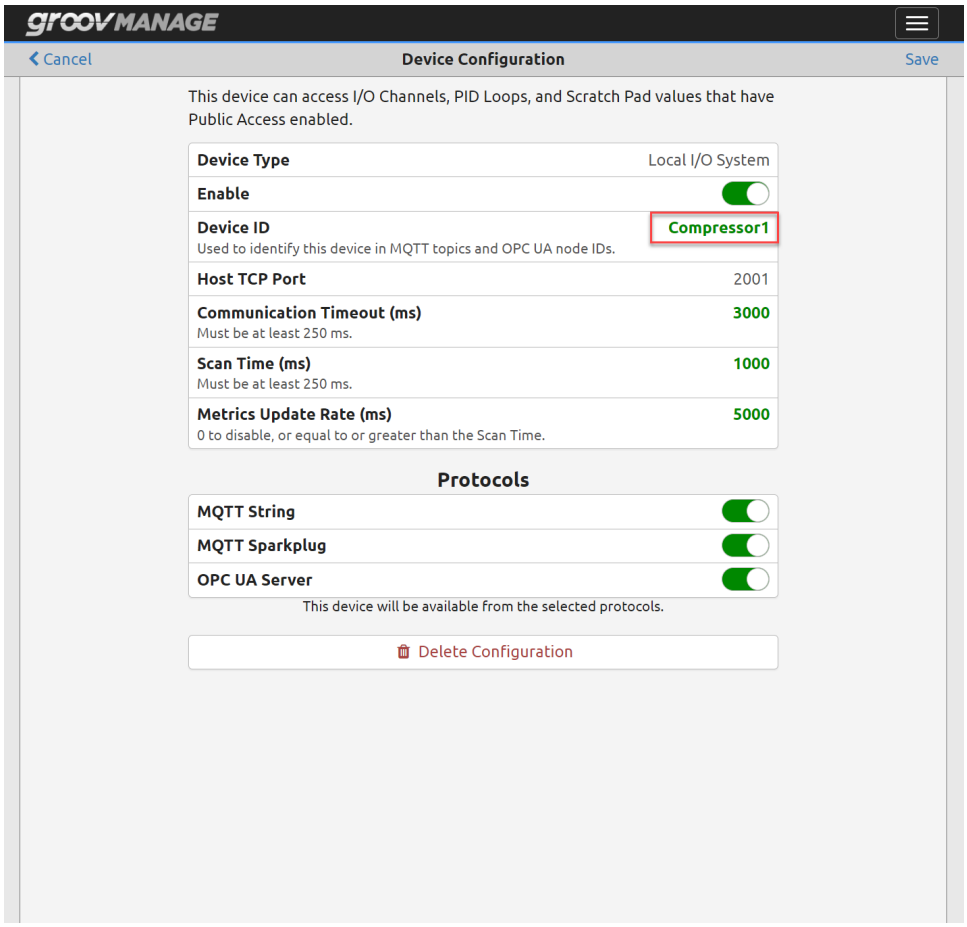
Configure the RIO EMU Data Service for OPC UA

1. Configure Data Services - Scanned Devices
    a. Click the “Data Service” button
    b. Click “Configure” at the top right corner

c. Under Scanned Devices, click "Add Local I/O System"



d. Next to Device ID, add a unique ID for this RIO EMU.
This Device ID should be the name of the load (e.g., Compressor1). No spaces are permitted; use underscores as necessary
Note: *The Device ID must be unique among all RIO EMUs in your system. Also, remember this Device ID name for later use in Ignition Designer*

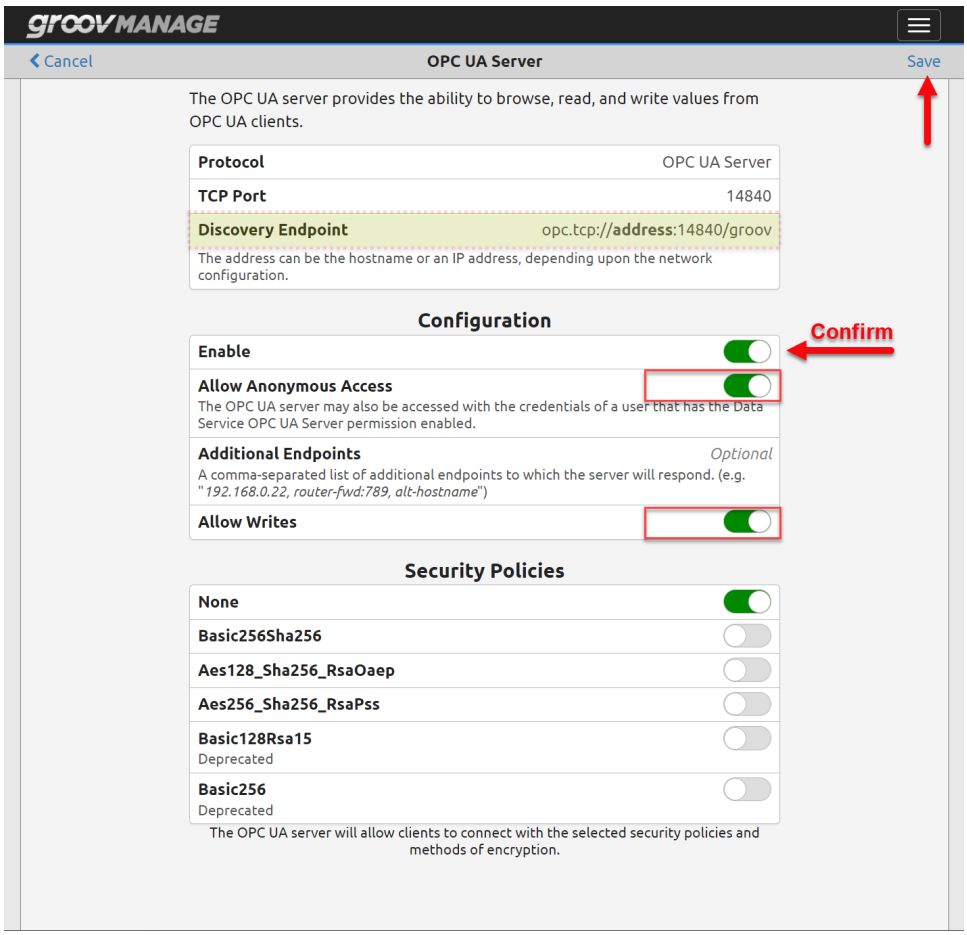


e. Click "Save" in the top right corner

2. RIO EMU OPC UA Configuration

a. From the Home Page, click "Data Service" button
b. Click the "Configure" link at the top right of the page
c. Click the "Add OPC UA" link
d. Under Configuration, ensure the Enable slider is ON
e. Note the Discovery Endpoint URL that reads:
   opc.tcp://**address**:14840/groov
   Your address will be your RIO EMU's hostname or IP address, and you'll need this information for configuration later in Ignition Edge.
f. Click the Allow Anonymous Access slider to ON (leave off for authenticated access only)
g. Click the Allow Writes slider to ON
h. If you wish to apply a Security Policy for encrypted access, select the appropriate option

i. Click “Save” in the top right of the page, then “Status” in the top left



3. RIO EMU commissioning is complete.

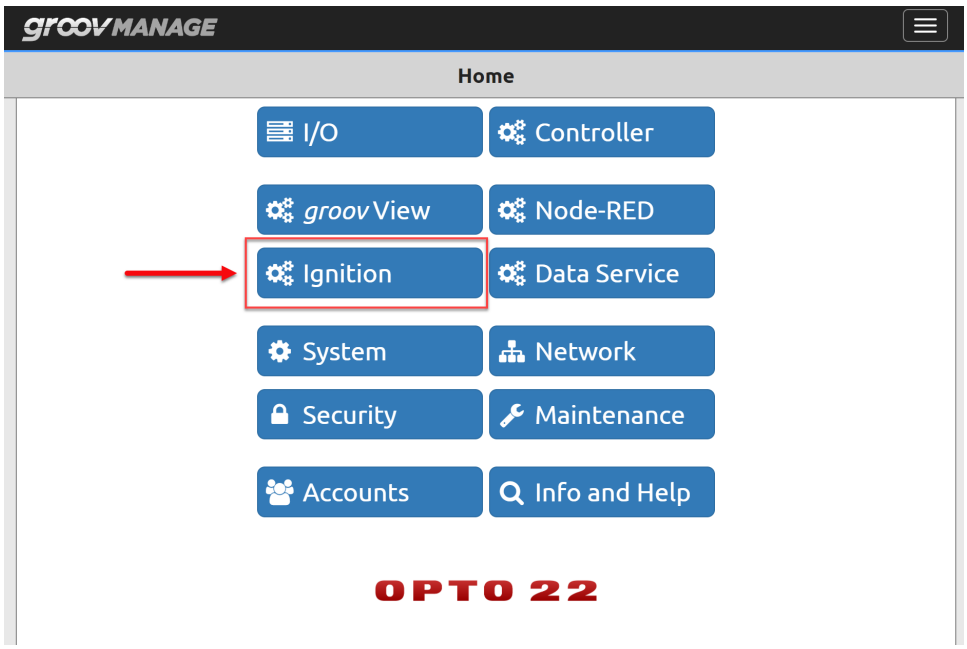
Configuring Ignition on EPIC and Installing Designer

Commissioning Ignition Edge (on *groov* EPIC PR2)

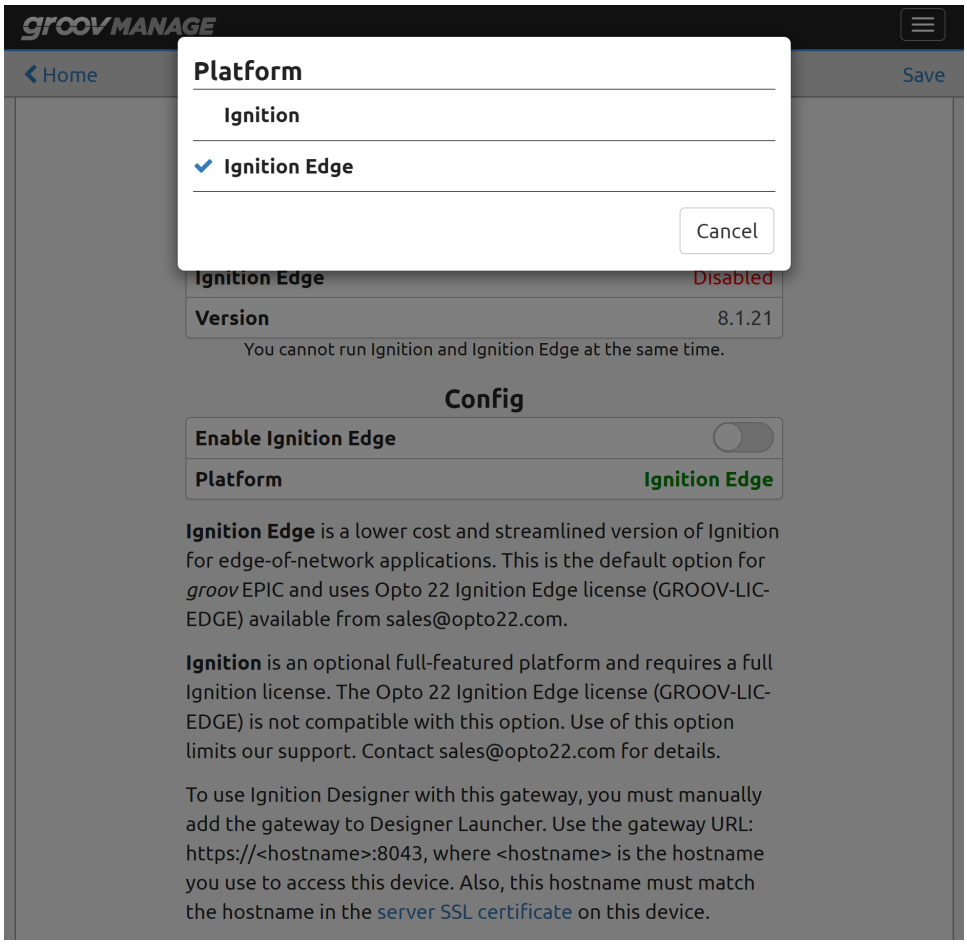
***Note**: These instructions assume you've already commissioned your groov EPIC. If you need assistance in commissioning your EPIC, see our free OptoU Training Course here:*
https://training.opto22.com/path/groov-manage-for-groov-epic/commissioning-groov-epic

1. From your browser, navigate to your groov EPIC's home page by typing in the hostname as follows: https://hostname-of-EPIC (e.g., https://epic-pr2-esab)
2. Log in with your EPIC's credentials

3. From the *groov* Manage Home page, click the blue Ignition button



4. Choose the Ignition Platform choice.
   For this exercise, we'll use Ignition Edge. Ignition Edge licensing is significantly lower cost than full Ignition, and provides most of the functionality required for both OPC UA and MQTT/Sparkplug.



5. Toggle the Enable Ignition Edge slider, and click Save



6. Follow the setup wizard by clicking OK



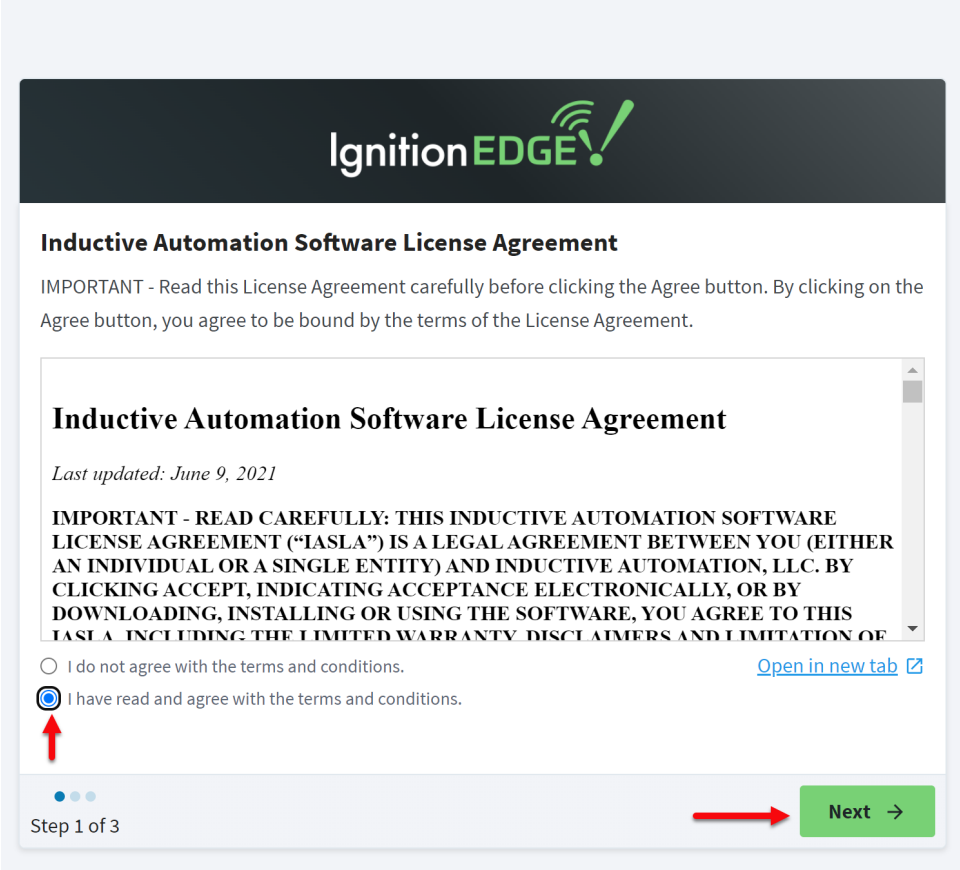
7. Wait while Ignition Edge is enabled. This can take a few minutes.

8. Once Ignition Edge has started, click the blue hyperlink on the resulting page that reads “Open Ignition Edge”:

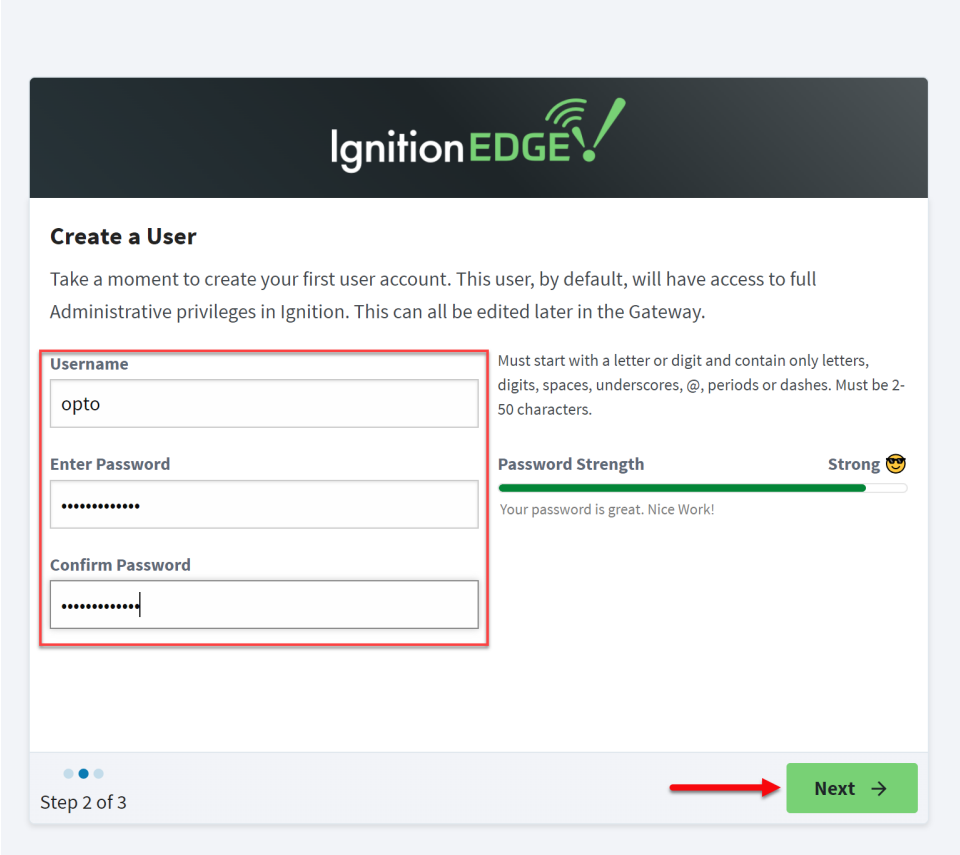


9. You’ll be directed to a new tab in your browser where the Ignition Edge gateway interface loads, and presented with 3 setup steps.

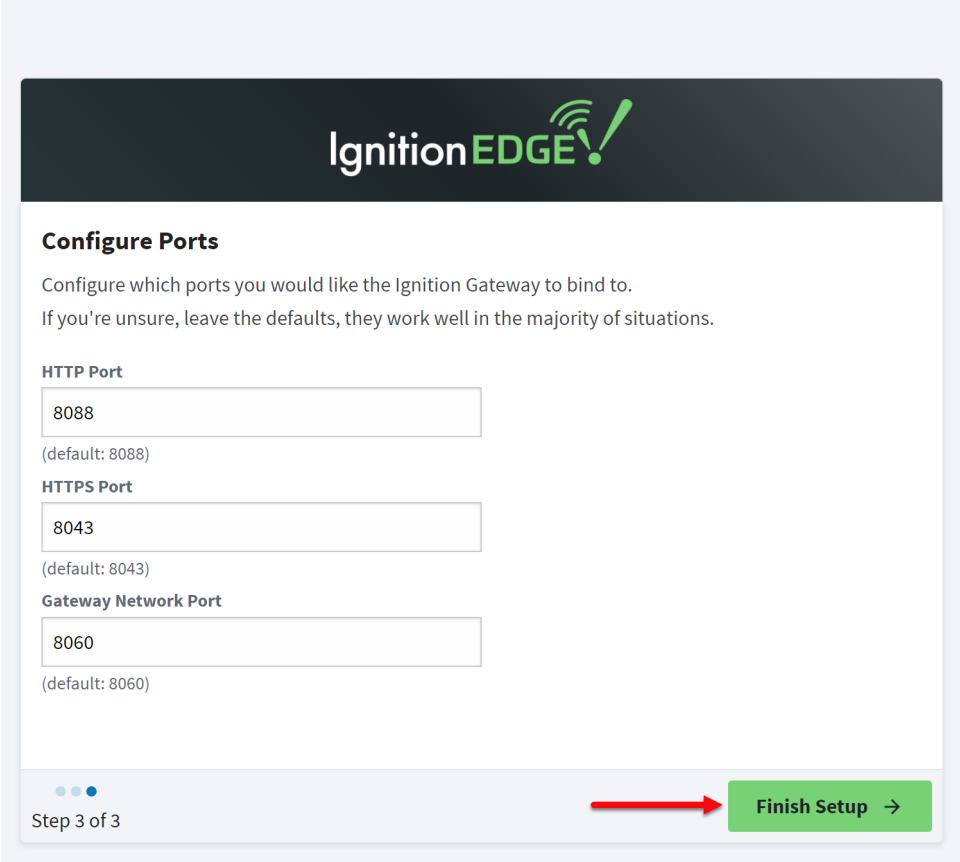
The 1st step is to accept the Inductive Automation Software License Agreement:



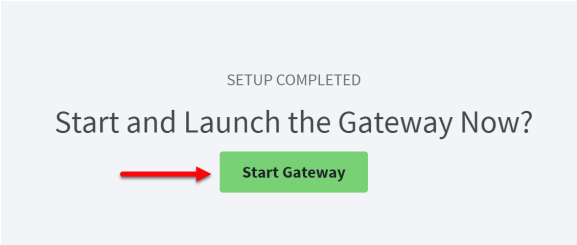
10. The 2nd Step is to enter in a Username and Password, then confirm your password.
***Note**: Store your username and password in a secure location. These cannot be recovered in the event you forget them.*



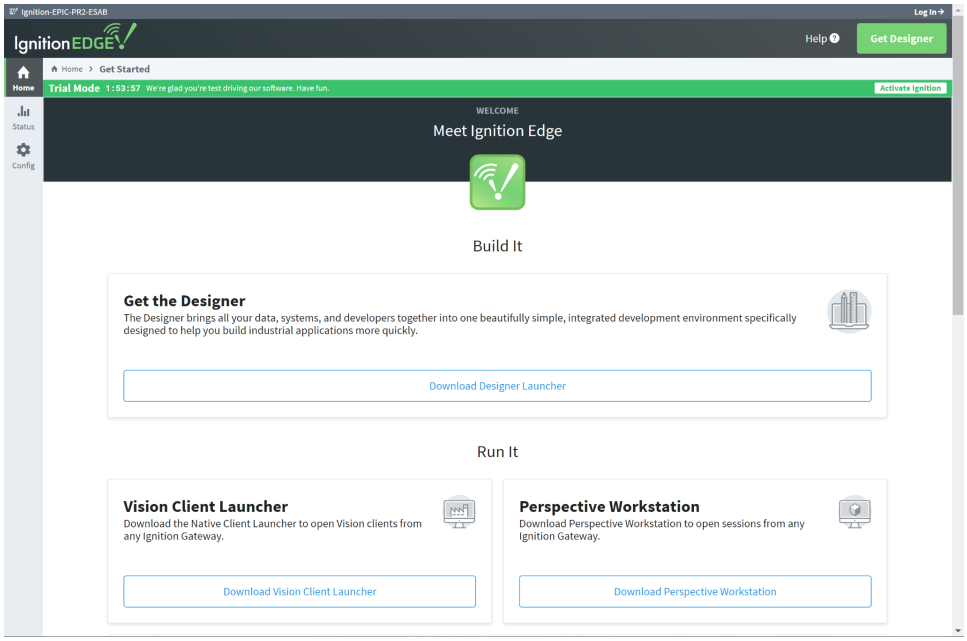
11. Configure your network ports for access. We recommend the defaults, so you may click Finish Setup to proceed.



12. Finally, Start and Launch the Gateway.
This will start the Ignition Edge Gateway. This can take up to 5 minutes to complete.

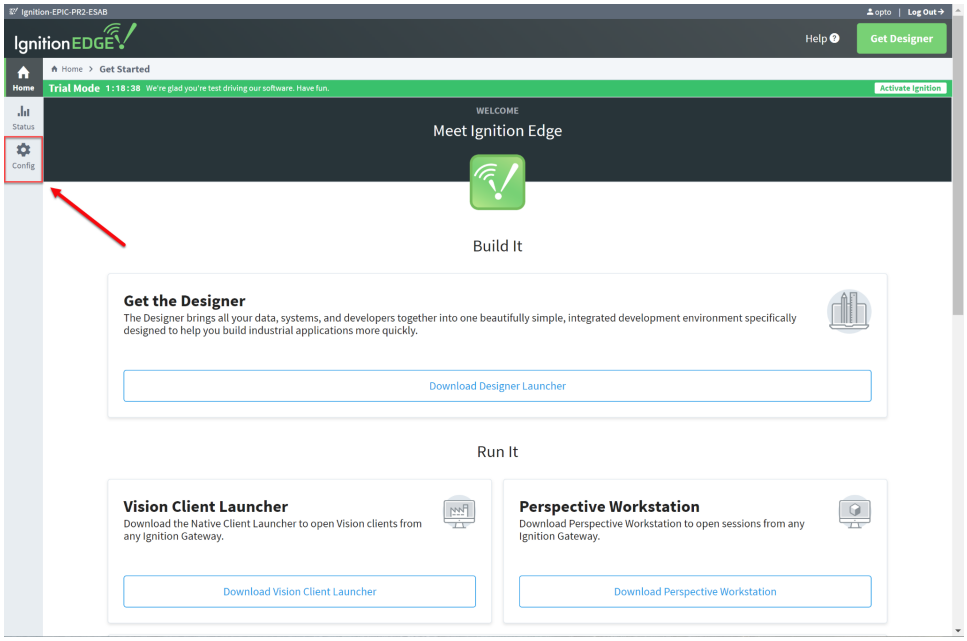


13. When complete, you'll see the Ignition Edge Gateway home page:

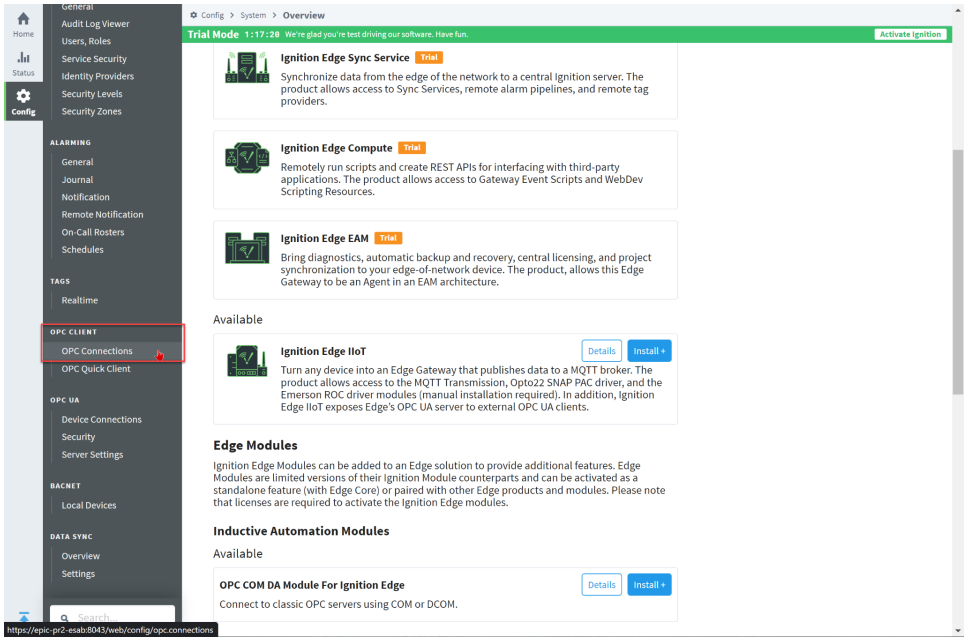


Ignition Edge OPC UA Configuration

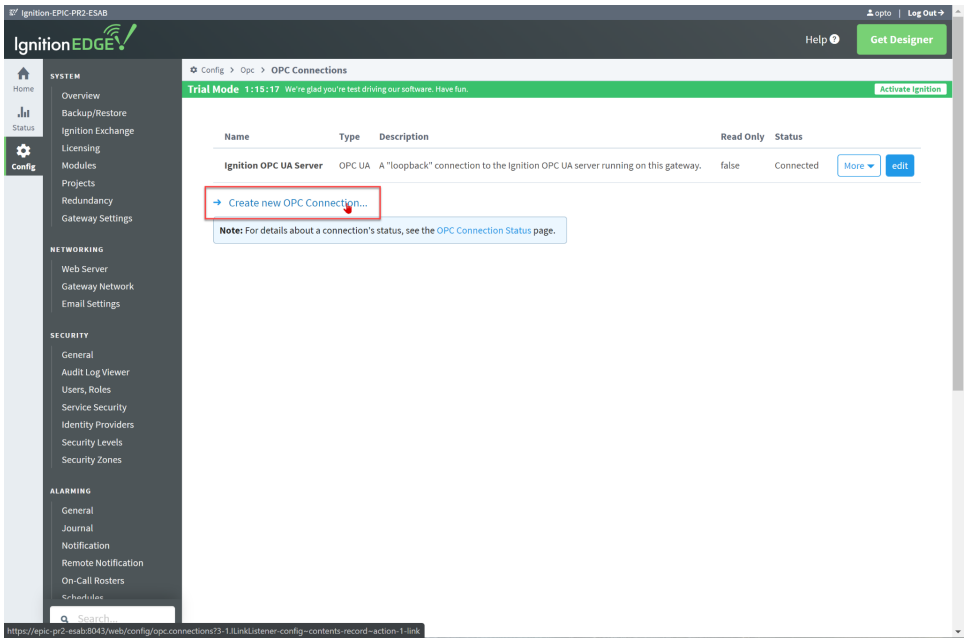
1. From the Home page, log in with your username and password by clicking on the “Log In” link at the top right corner
2. On the left of the screen, click the Config link



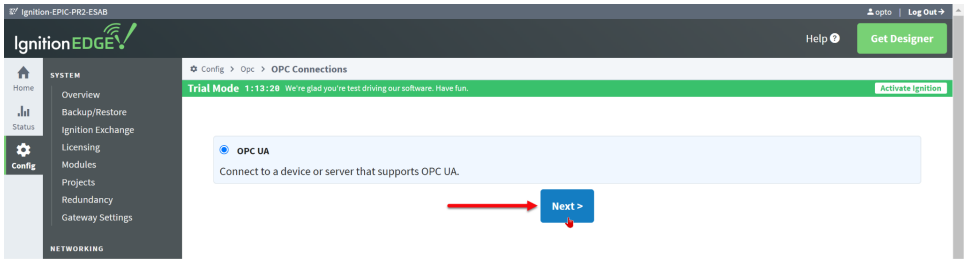
3. Scroll down on the resulting page, and click OPC Connections under the heading OPC CLIENT



4. Click the blue link that reads “Create new OPC Connection…”

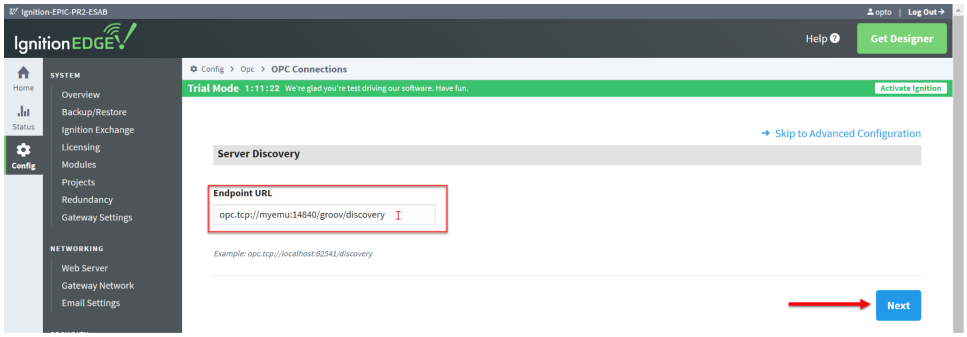


5. Follow the wizard by clicking “Next”

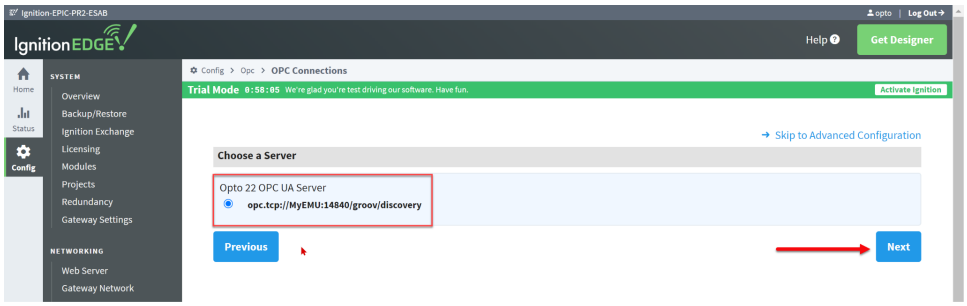


6. In the Endpoint URL box, enter the path to your RIO EMU’s OPC UA Server (noted from RIO EMU commissioning above):
`opc.tcp://myemu:14840/groov/discovery`

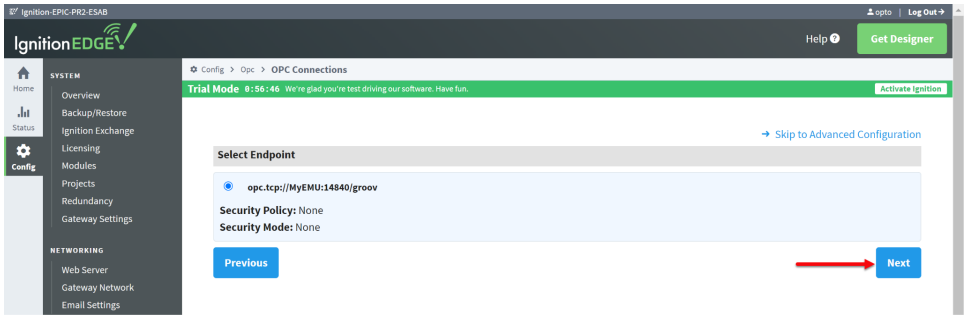
Then, click Next



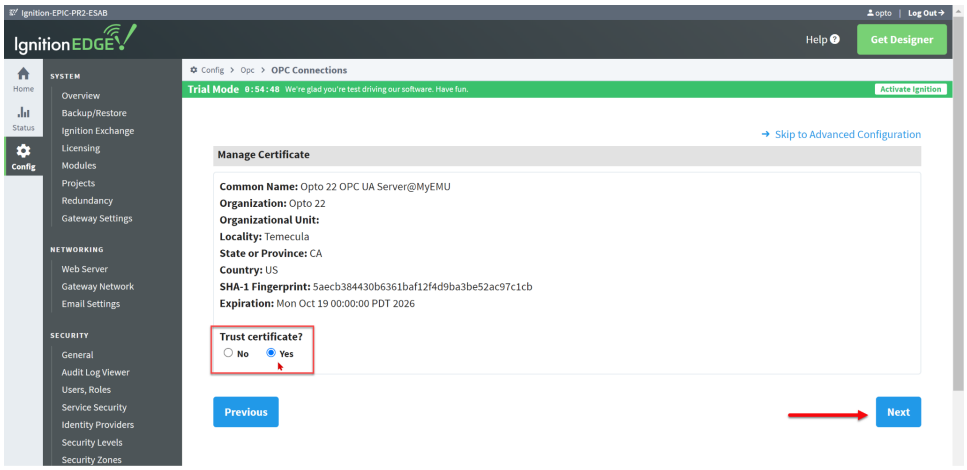
7. Confirm the server address, and click Next



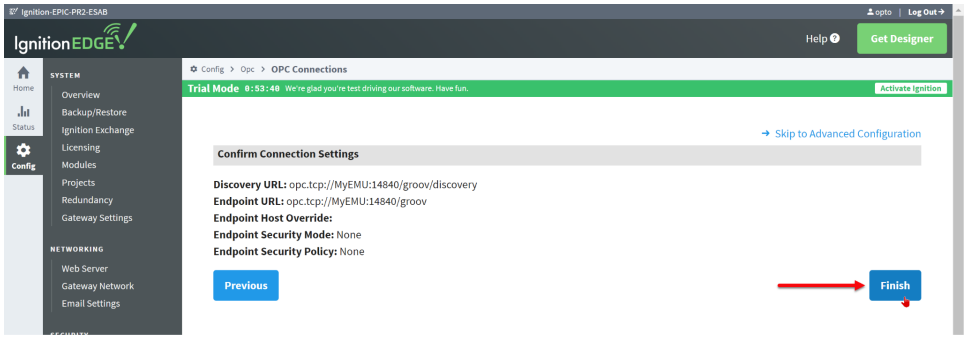
8. Click Next again after confirming Endpoint



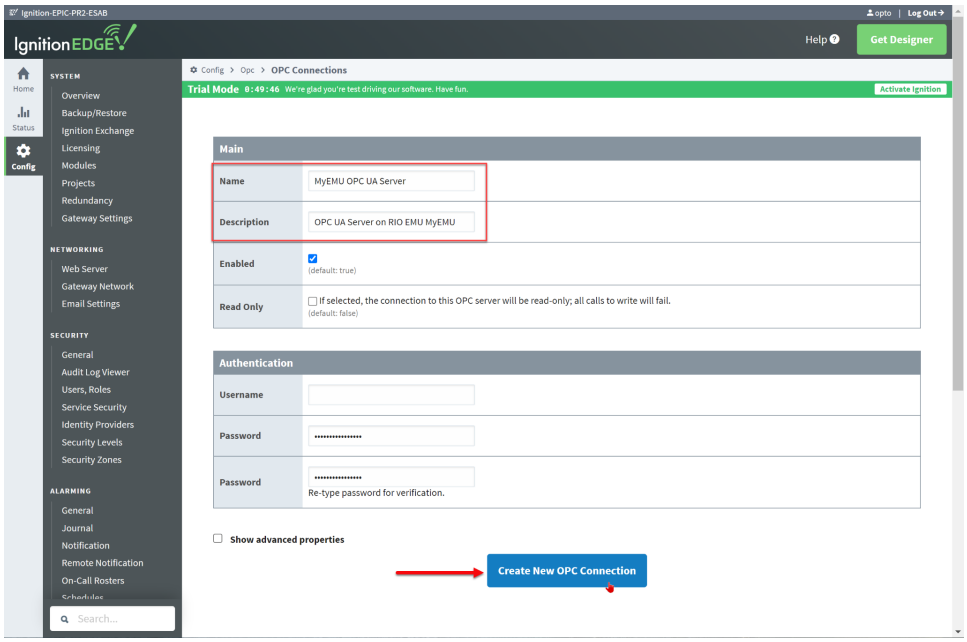
9. Select “Yes” for the prompt that reads “Trust certificate?”, and then click Next



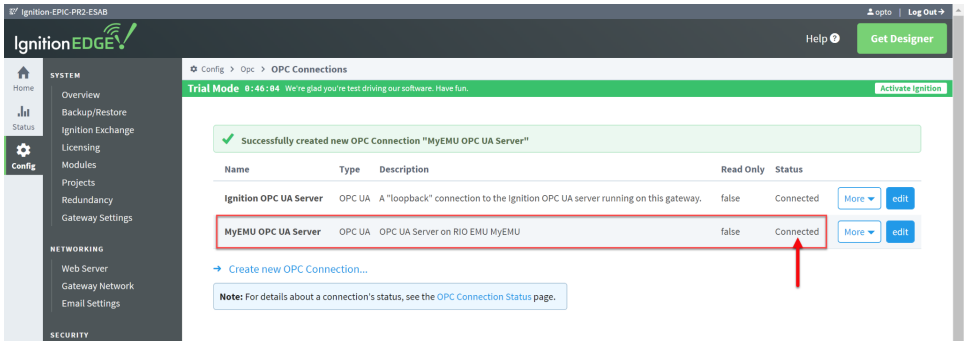
10. Confirm your Connection Settings, and click Finish



11. In the Name field, enter a name for this OPC UA connection. We recommend using the hostname of the RIO EMU in the name, for example:
MyEMU OPC UA Server
Add a Description if you like, and click the blue button Create New OPC Connection.
Note: *Remember this name for use later when instantiating UDT in Designer*

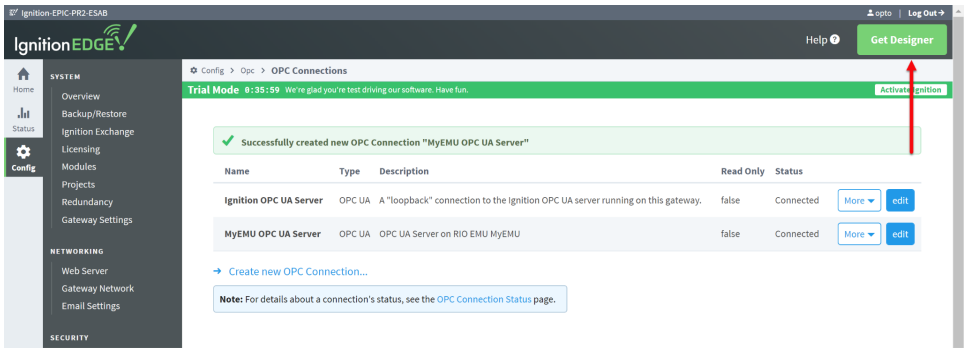


12. On the resulting page, you should see your new OPC UA connection, along with “Connected” under the Status column

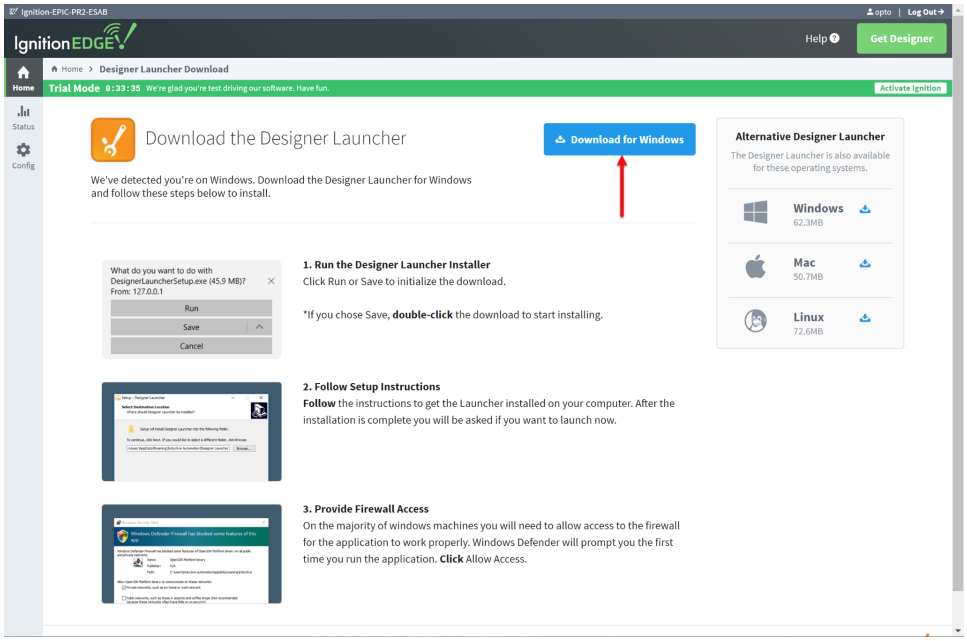


Ignition Edge Designer install and configuration

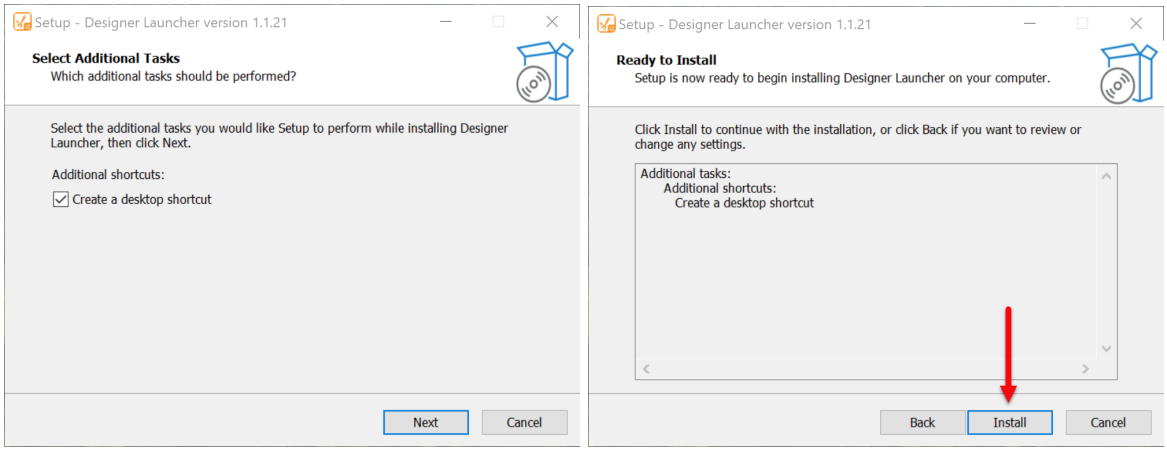
1. At the top right of the page, click the green button labeled “Get Designer”



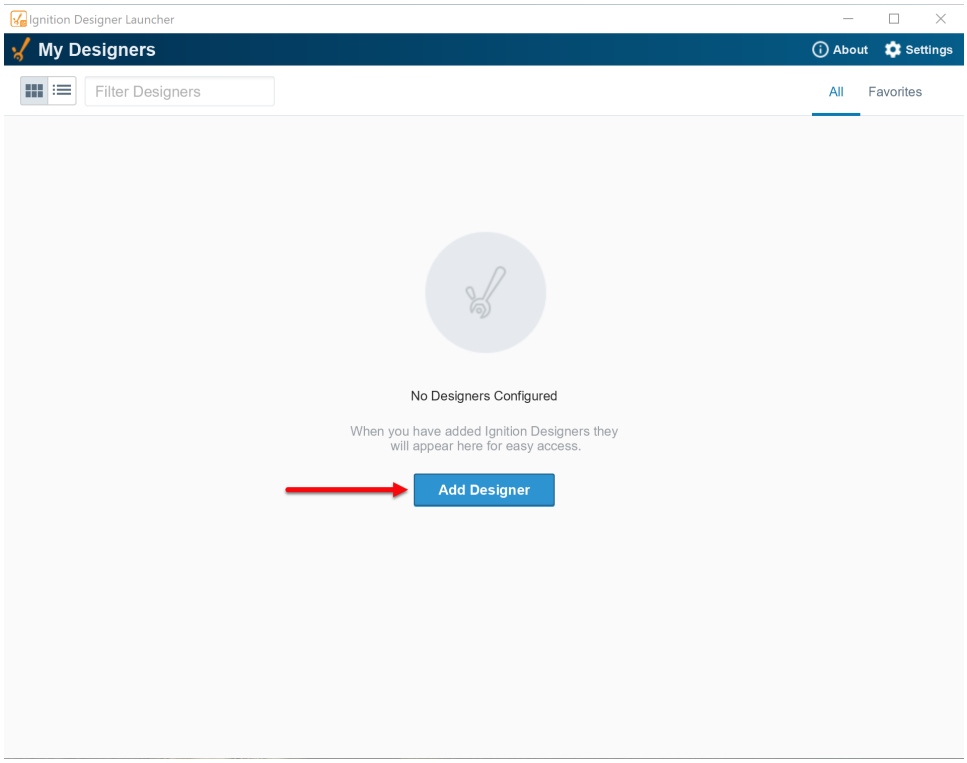
2. Click the blue button to download Designer for your operating system to your computer



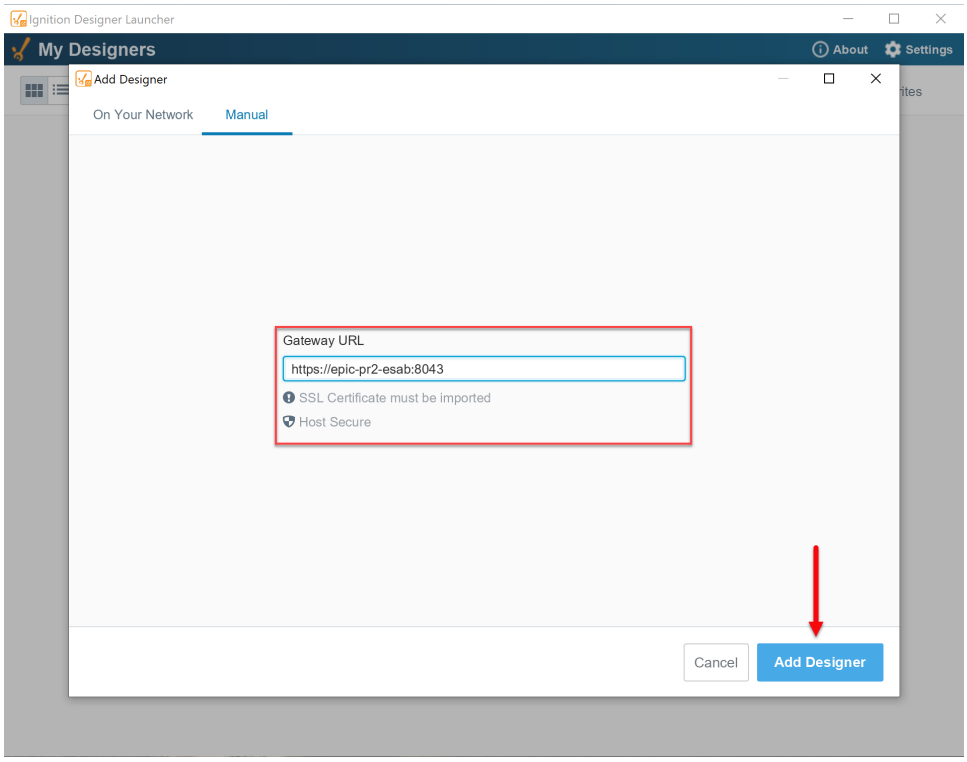
3. Locate the file you downloaded on your PC, and double click it to install. You should see the following screen, then click Next, and then Install, and finally Finish



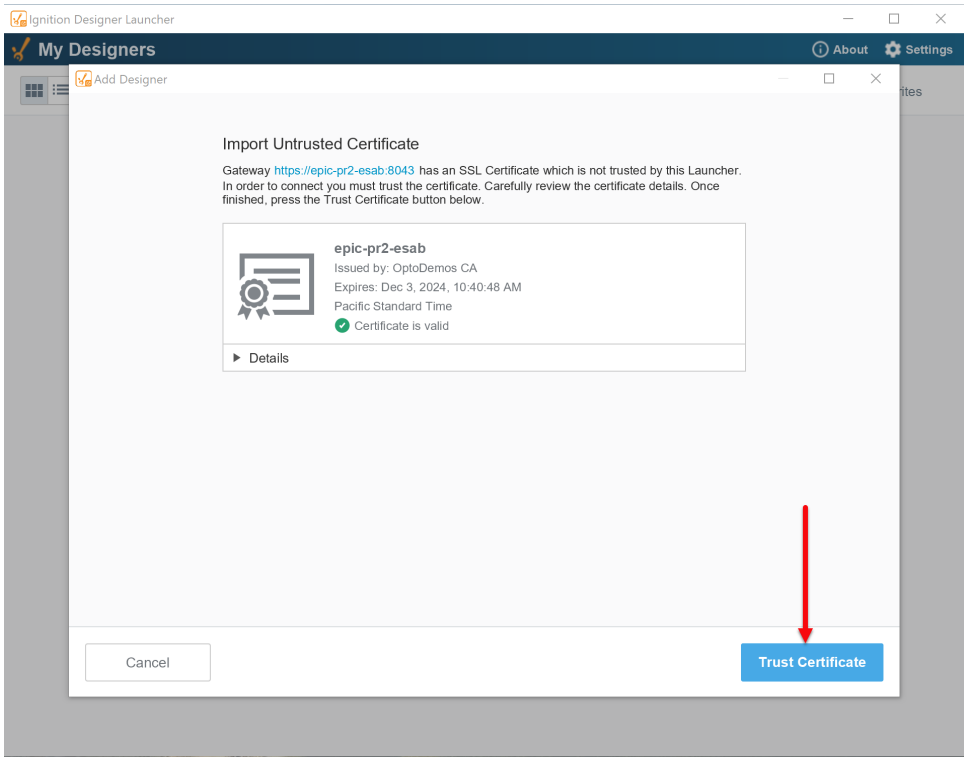
4. With Designer running, click the blue Add Designer button



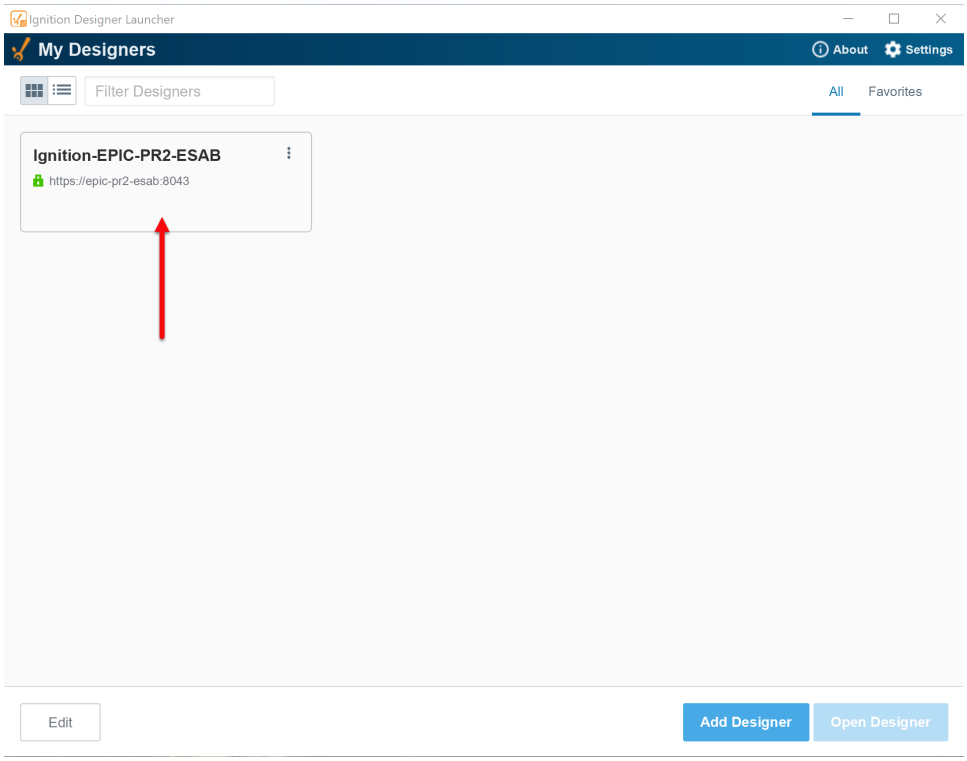
5. Click on “Manual” near the top of the page, and enter your EPIC’s hostname followed by port number 8043 in the Gateway URL box (e.g., https://epic-pr2-esab:8043), then click Add Designer



6. Click the blue Trust Certificate button to import the certificate



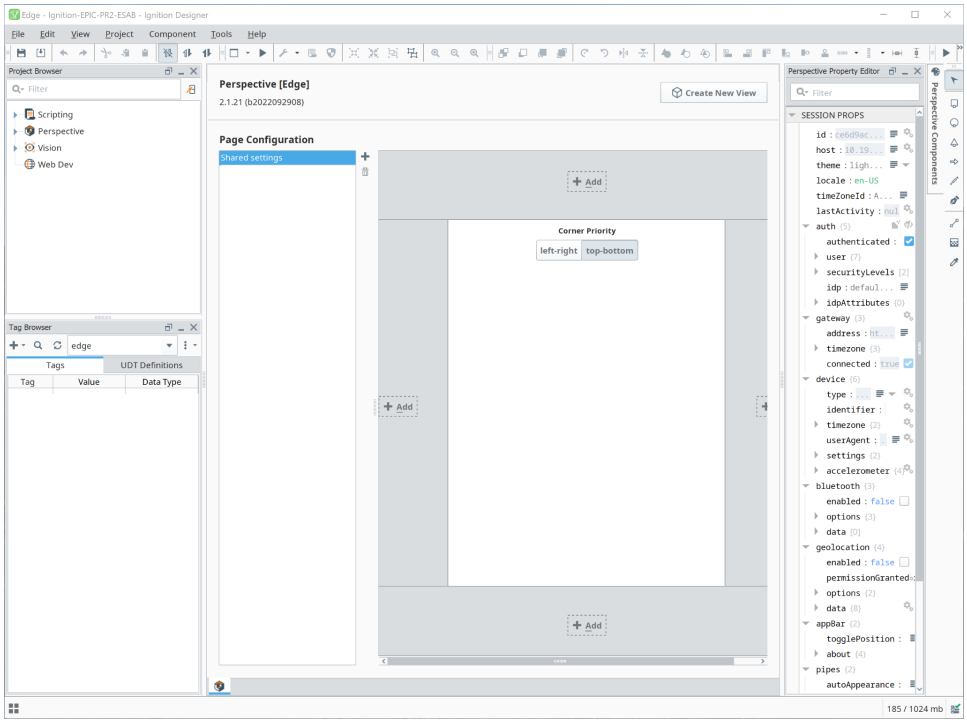
7. On the resulting screen, double-click on your configured Designer



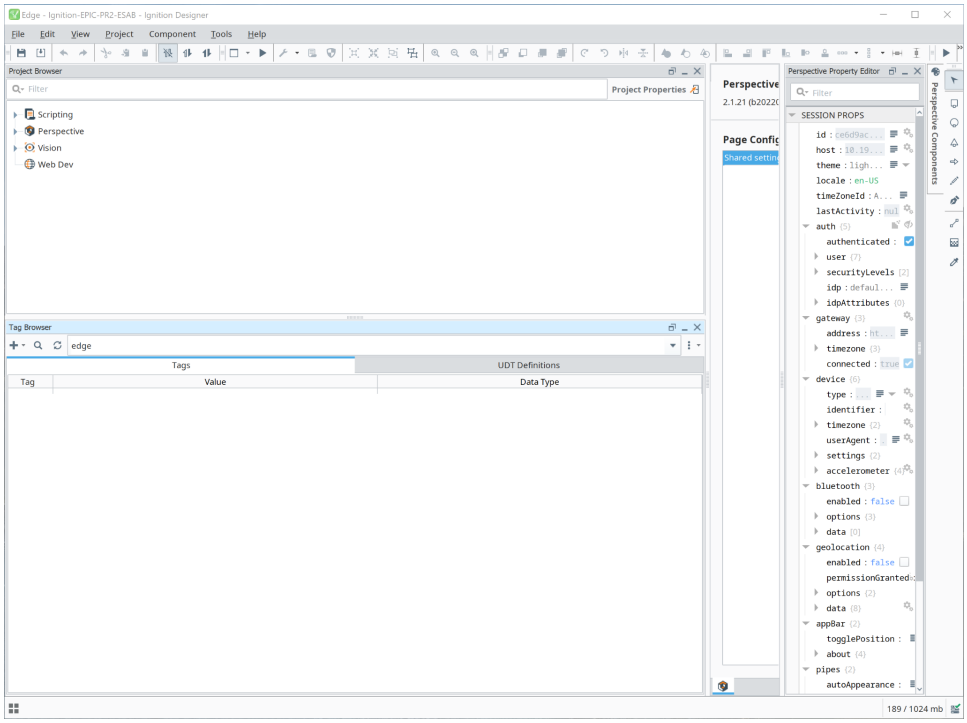
8. Designer will begin to load, then prompt you for your username and password. Enter them and click Login



9. Once loaded, your Designer application should appear like so:

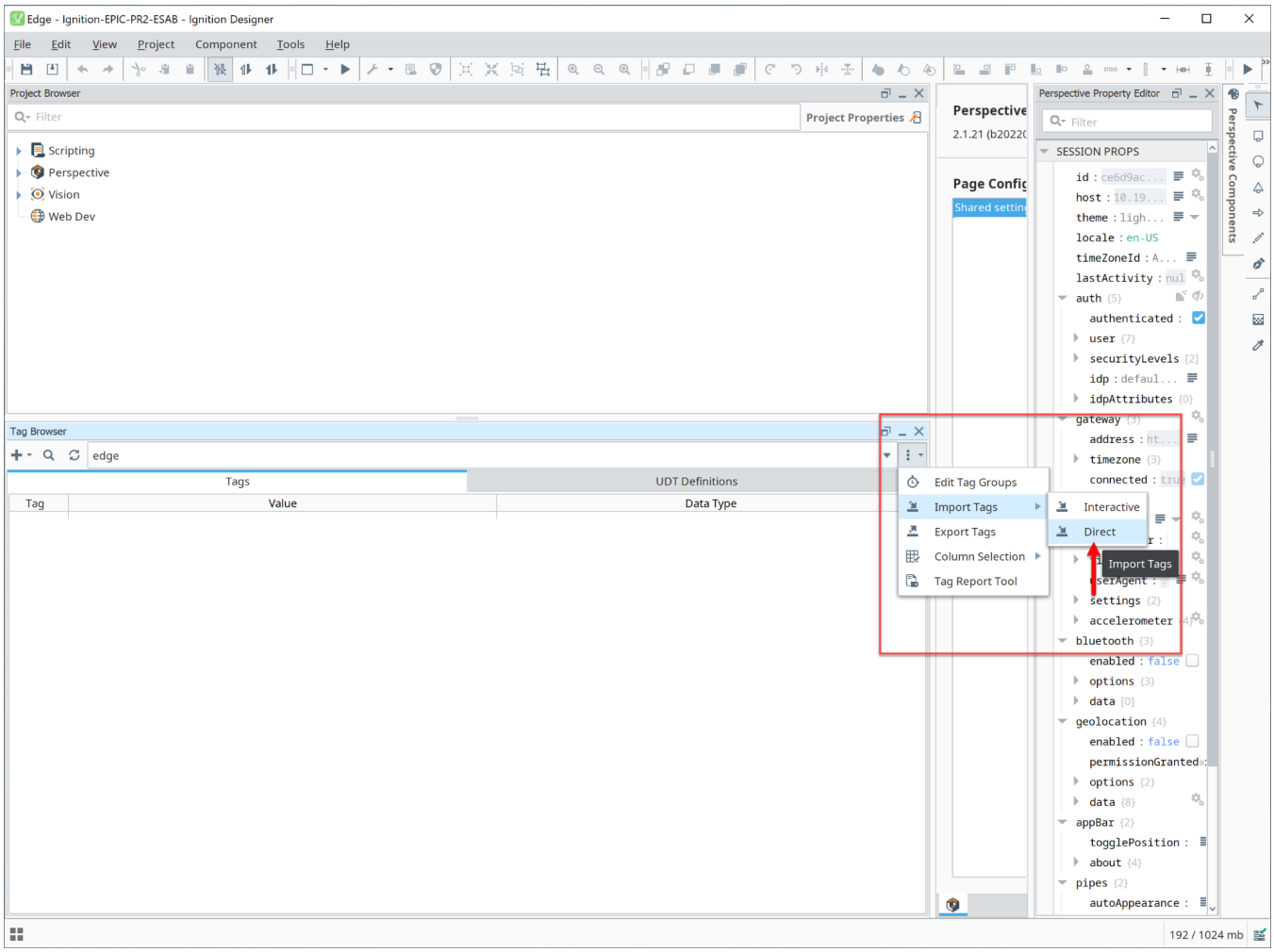


10. Most of the configuration we'll do in Designer is in the Tag Browser. We recommend adjusting the panes in Designer by dragging the Tag Browser's right pane divider to the right, like so:

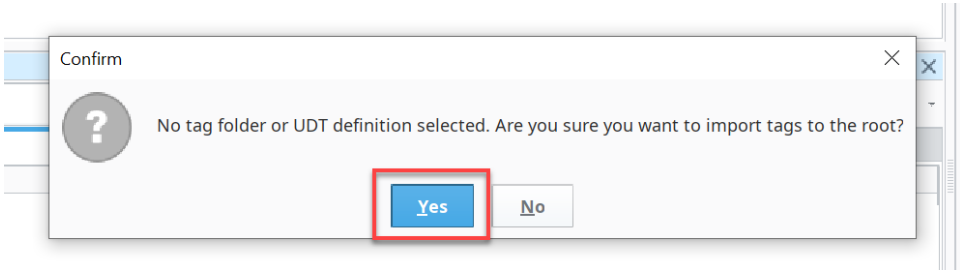


Importing and Configuring the Energy Monitoring UDTs in Ignition

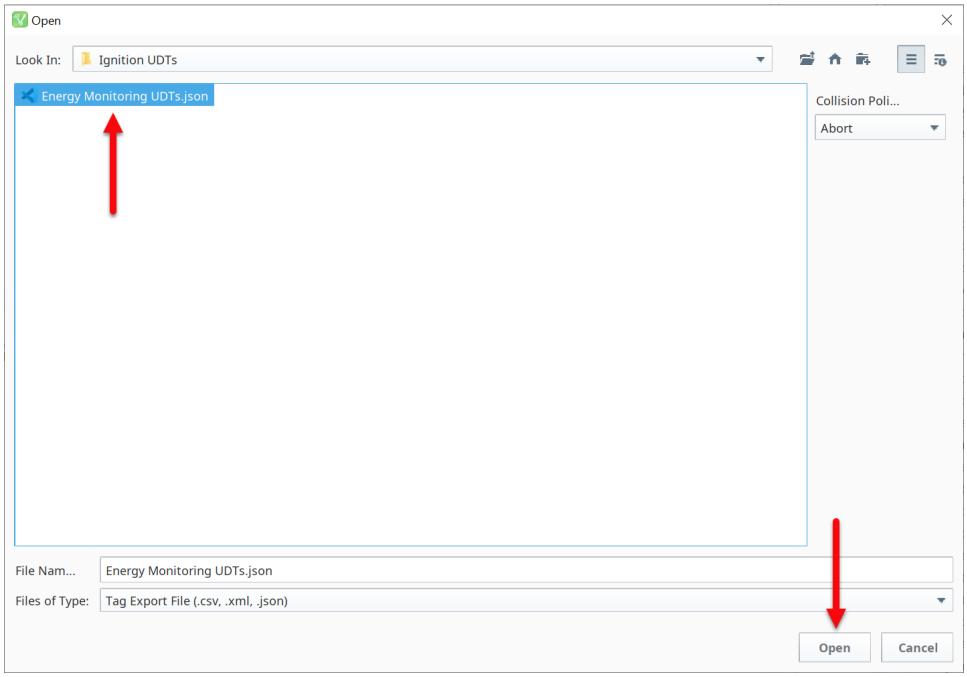
1. From within the Tag Browser pane, click on the 3 vertical dots at the top right, and choose Import Tags, Direct



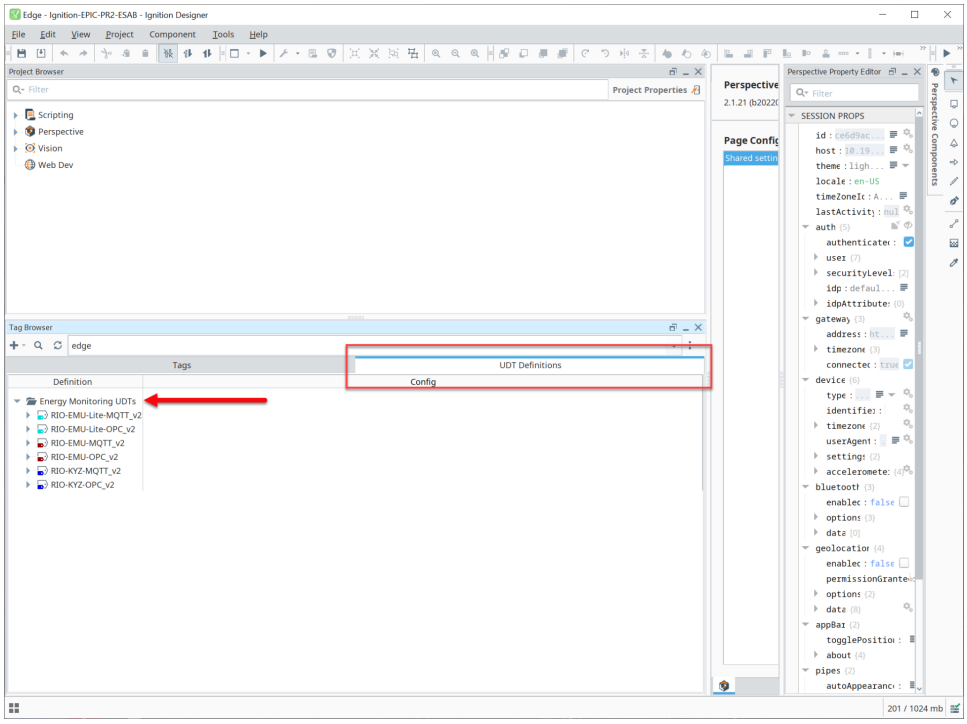
2. Choose Yes in the resulting dialog



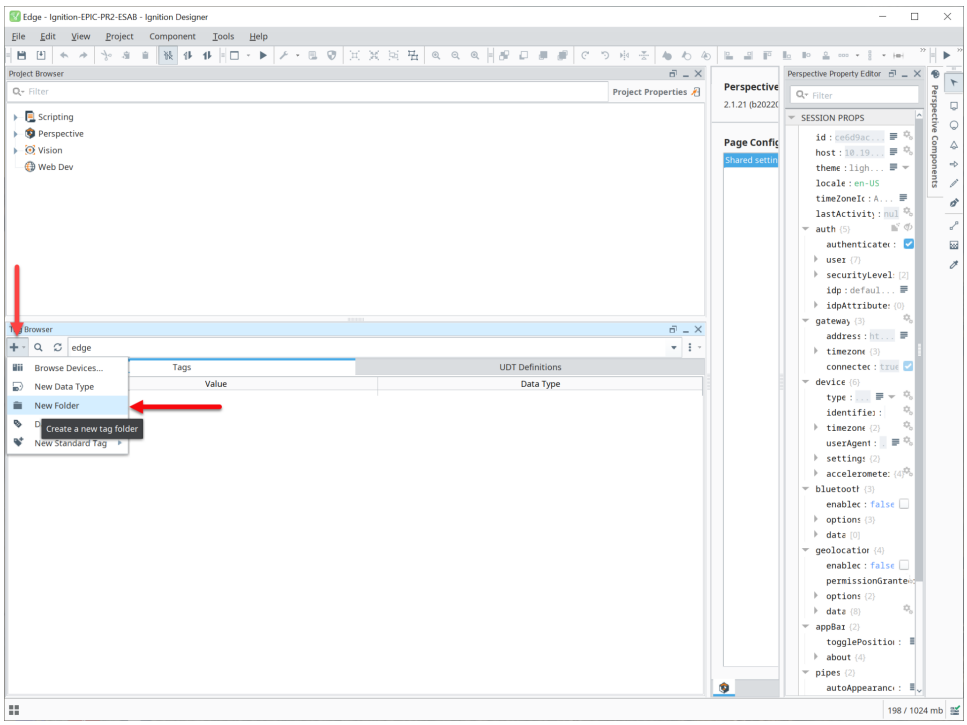
3. Navigate to where you've downloaded the Energy Monitoring UDTs.json file, and select it, and click Open



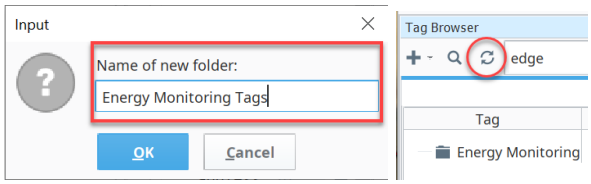
4. Click on the tab labeled "UDT Definitions" to see your imported UDTs, then expand the folder



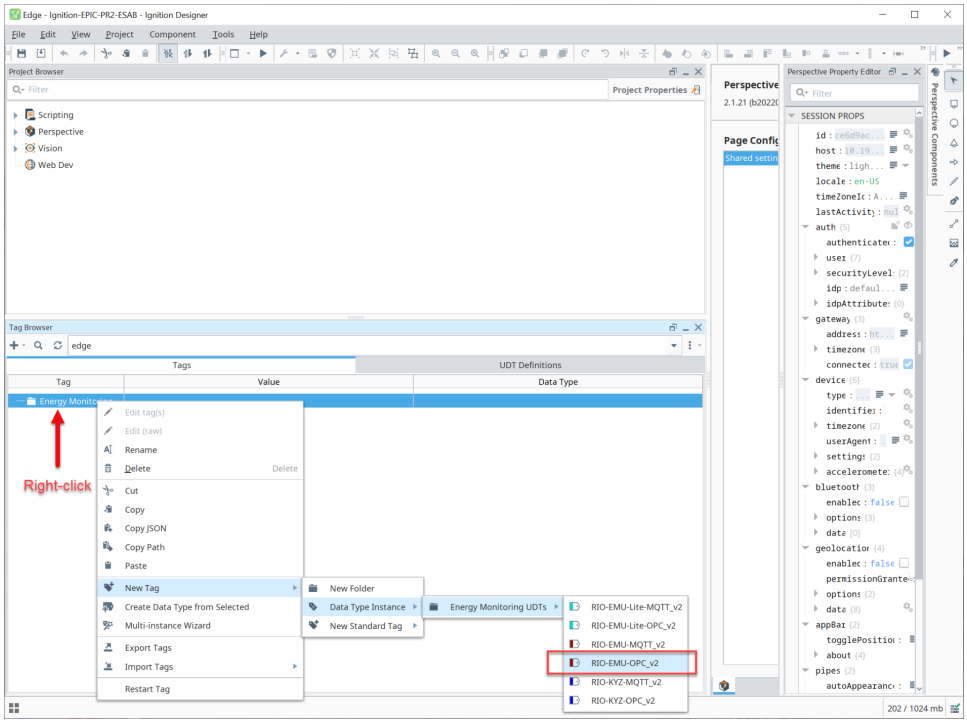
5. Switch back to the Tags tab, click the + sign at the top left of the Tag Browser pane, and choose New Folder



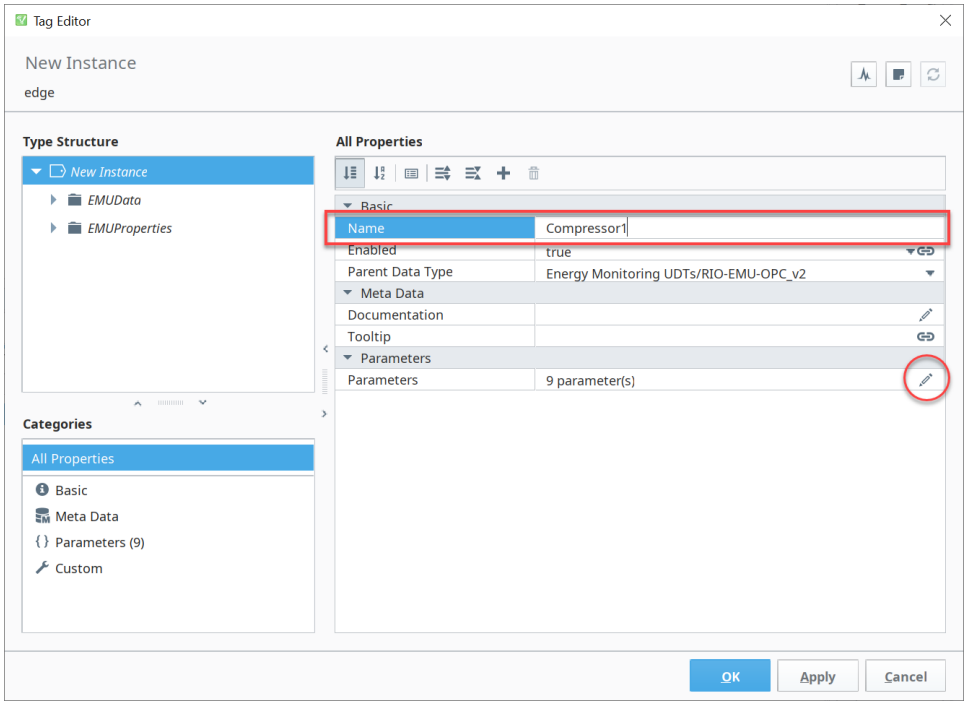
6. Name your new folder “Energy Monitoring Tags” and click OK. If necessary, click the Refresh icon in the Tag Browser pane to see your tags or folders



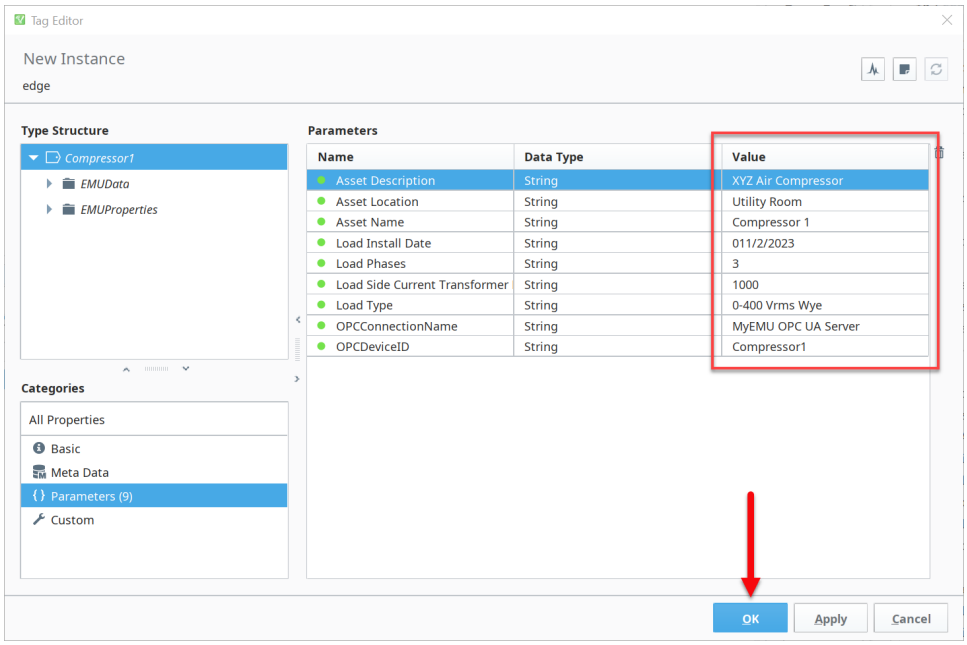
7. Right click the Energy Monitoring Tags folder, click on New Tag > Data Type Instance > Energy Monitoring UDTs > RIO-EMU-OPC_v2 to create an instance from the UDT:



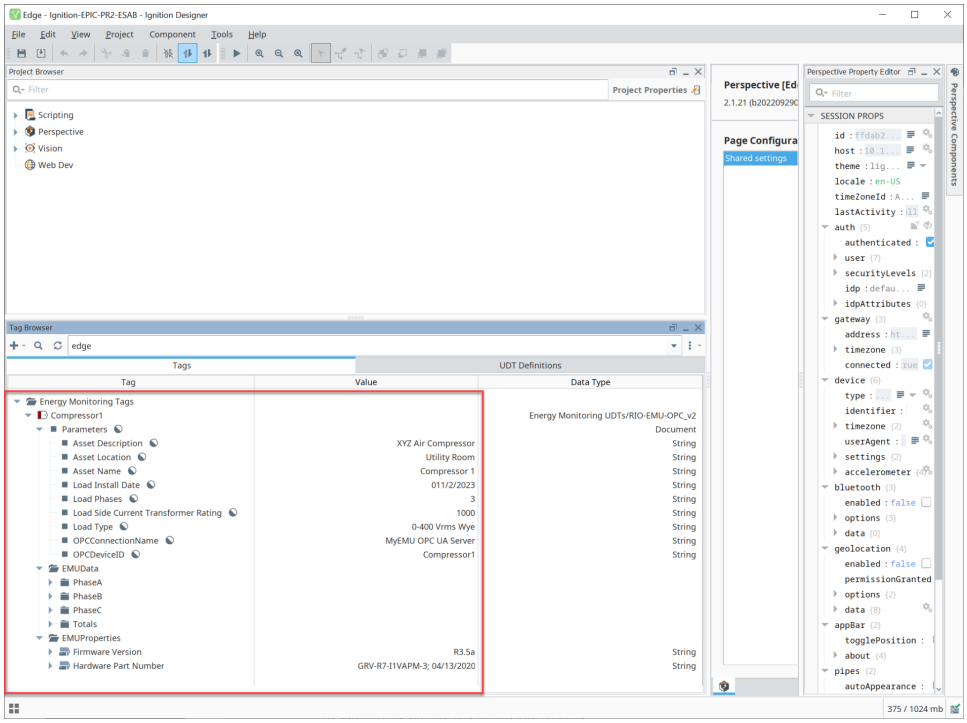
8. In the resulting Tag Editor dialog, name your instance. Ideally, name your instance as the Device ID noted earlier (e.g., “Compressor1), then click the pencil icon next to Parameters:



9. Here, you’ll enter the parameters for this UDT instance related to this RIO EMU. This information comes from how you will model this data along with parameters about the specific load as configured in the RIO EMU commissioning steps:
    a. **Asset Description:** Describe the asset being measured
    b. **Asset Location:** Where the asset is located in the building or site
    c. **Asset Name:** The human-readable name of this asset
    d. **Load Install Date:** The date this asset was installed
    e. **Load Phases:** The number of phases being measured (from *groov* Manage on the RIO EMU)
    f. **Load Side Current Transformer:** The load side rating of the installed CTs (from RIO EMU)
    g. **Load Type:** The voltage range and Delta/Wye configuration of the install (from RIO EMU)
    h. **OPCConnectionName:** The OPC UA Connection name used previously (from Ignition Gateway OPC Connection configuration)
    i. **OPCDeviceID:** The Device ID configured previously with *groov* Manage (from RIO EMU)
10. Review your configuration, and click OK



11. Click the Refresh button, and expand your Energy Monitoring folder to see your live values from the RIO EMU



12. You have successfully imported your UDTs, instantiated a UDT into a modeled asset, and can now develop your Ignition application. In the next steps, we'll configure sending your modeled data up to Ignition Cloud Edition.

Configuring Ignition Edge to Publish your UDTs to Ignition Cloud Edition

Now that we have live, modeled energy data for your asset in Ignition Edge, the next steps will be to publish that data to Ignition Cloud Edition. Note that these steps are very similar to publishing your energy data to an on-premise Ignition Gateway as well.

These steps assume that Ignition Cloud Edition is already up and running in your chosen cloud environment (AWS or Azure). Ignition Cloud Edition includes all the necessary Ignition Modules to perform this task.

If you're publishing instead to an on-premises Ignition Gateway, that Gateway will need the following Ignition Modules installed and configured:

- MQTT Distributor - an MQTT broker/server embedded into Ignition. Alternatively, any MQTT 3.1.1. Compliant broker installed on the network will work.
- MQTT Engine - This module will subscribe to the UDTs from the MQTT broker, and populate your chosen Tag Provider with the live data

In either case, the primary MQTT configurations needed to complete these steps are a valid URL endpoint to the broker (whether that's MQTT Distributor or any 3.1.1. Compliant broker), and a configured MQTT user/pass credentials for any clients (including Ignition Edge operating as an MQTT Client).

For this example, we're using Ignition Cloud Edition with the MQTT Distributor module installed at the URL endpoint: https://ice.groov.com:8883

The MQTT Client credentials for this example are:

- **Username**: IgnitionEdge_Client
- **Password**: opto22!!

The MQTT Namespace for this example is:

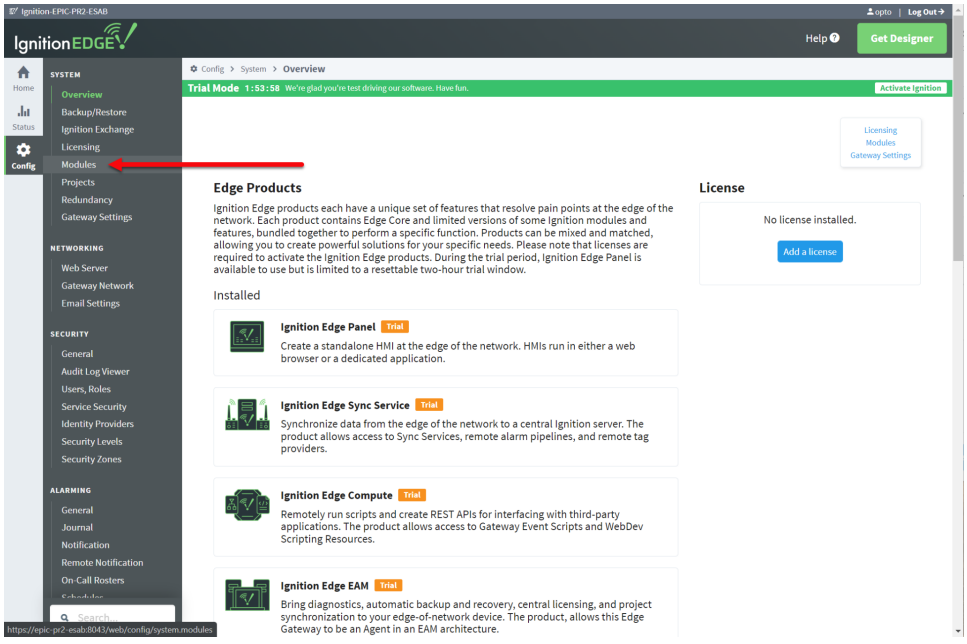
- **Group ID**: OptoFactory
- **EdgeNode ID**: EPIC-PR2-ESAB (the hostname of the PR2 running Ignition Edge)
- **Device ID**: Energy

The Primary Host ID for this example is:

- **Primary Host ID**: ICEPrimaryHost (the Primary Host ID configured in MQTT Engine on Ignition Cloud Edition)

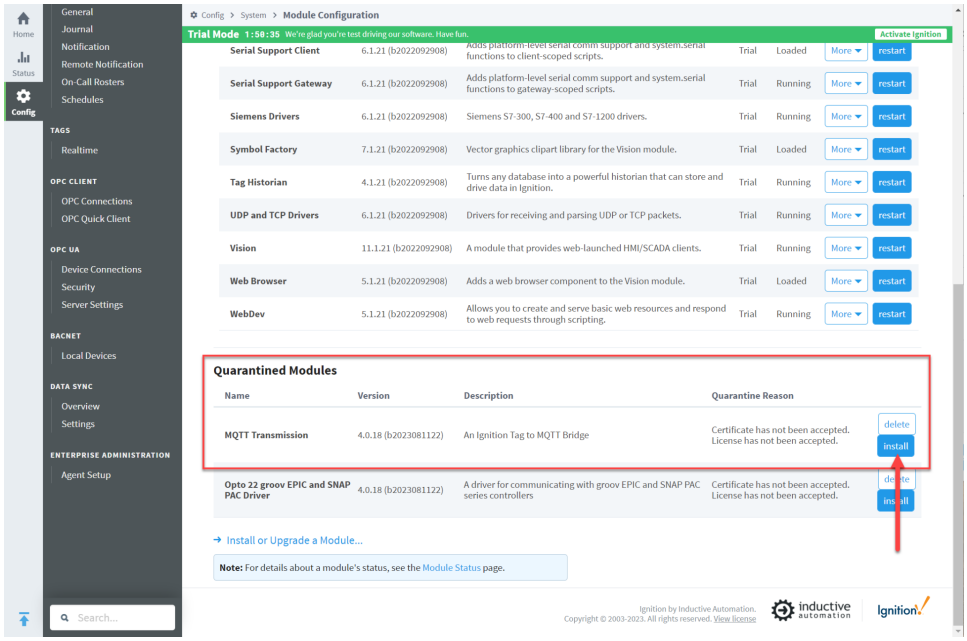
Let's get started.

1. Open up and log in your Ignition Edge Web Gateway from groov Manage (e.g., https://EPIC-PR2-ESAB > Ignition > Open Ignition Edge)
2. Click the Config icon in the left navigation
3. Click on the menu item "Modules" in the left navigation



4. In the resulting Module Configuration page, scroll down to the bottom of the page under the heading Quarantined Modules, and next to MQTT Transmission, click Install
   ***Note**: This is a one-time step on a freshly configured EPIC/Ignition Edge; if your page already shows this*

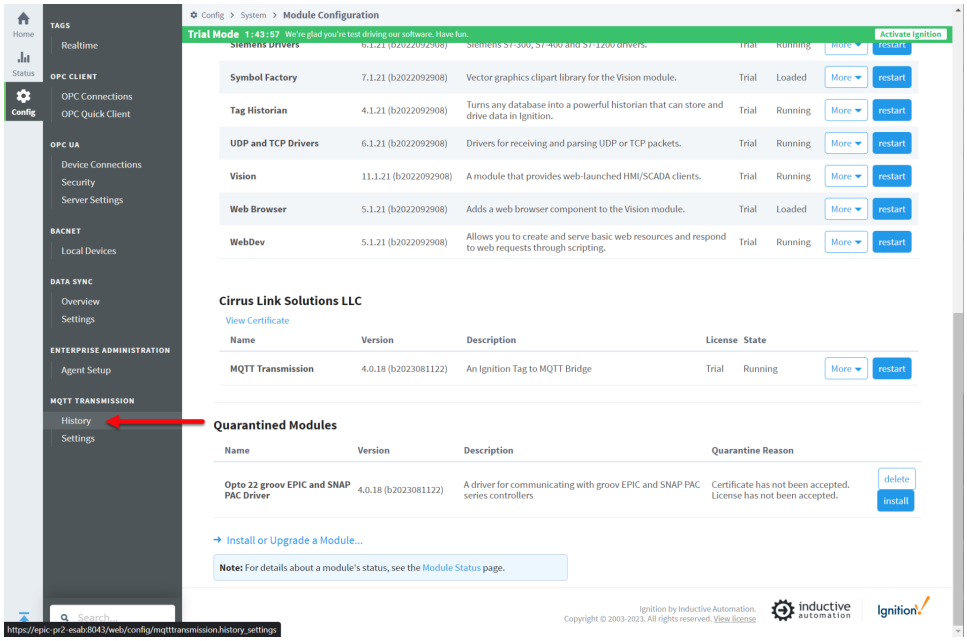
module is installed and running, you can move on.



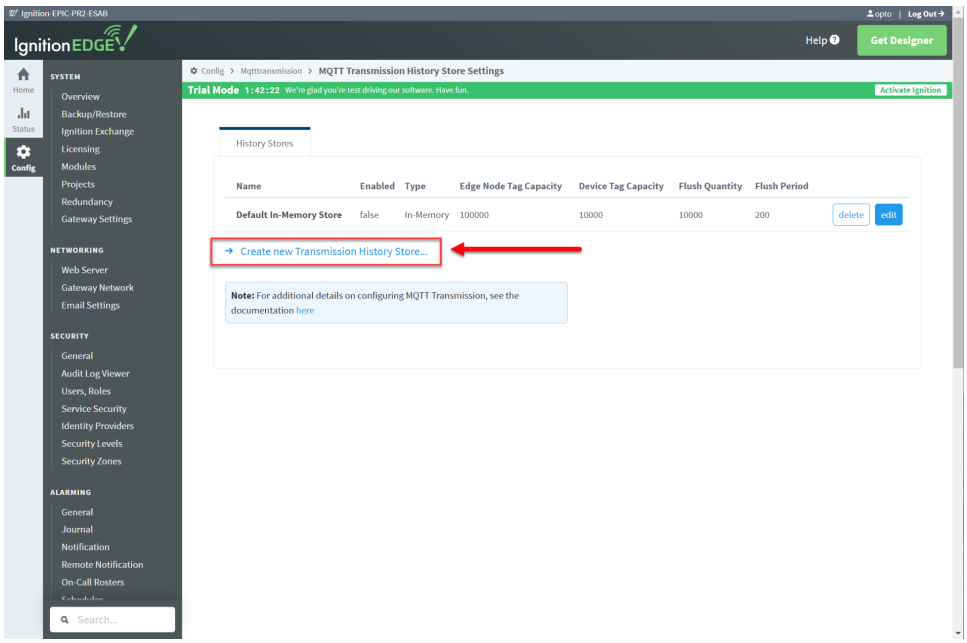
5. Follow the prompts to install the module, accept the terms of the License Agreement, check the box to add the certificate, and install the module.
6. Back on the Module Configuration page, scroll down to MQTT Transmission, and confirm the state “Running”



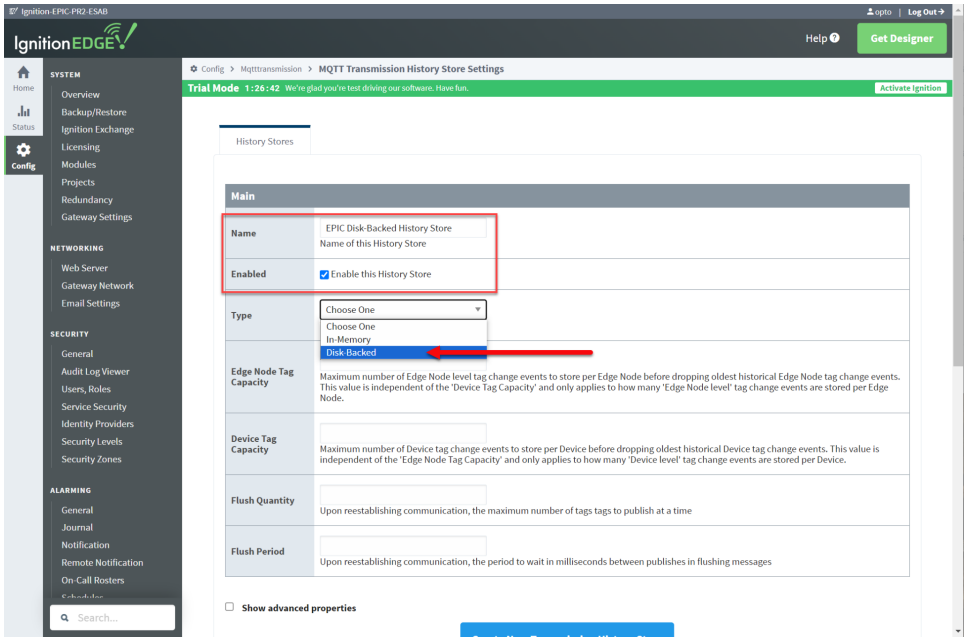
7. In the left nav, locate the heading MQTT Transmission, and click on History



8. Click on the blue link named Create new Transmission History Store…

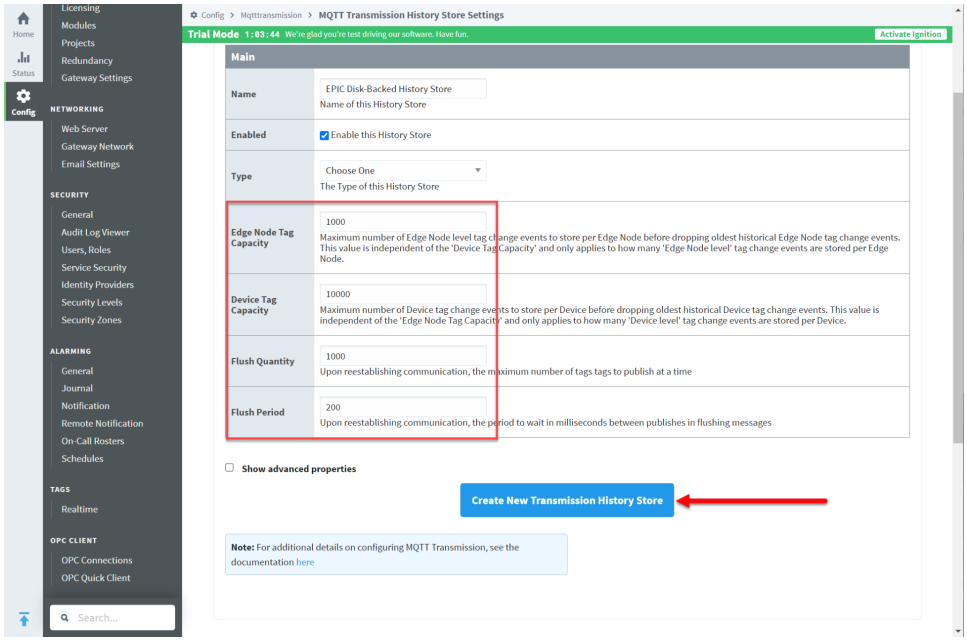


9. Give your History Store a name (e.g., EPIC Disk-Backed History Store), ensure the Enabled check box is checked, and for Type, choose Disk-Backed. This ensures your History Store will write store & forward data to on-board SSD in the event your connection to the broker is interrupted or your Primary Host is unreachable.

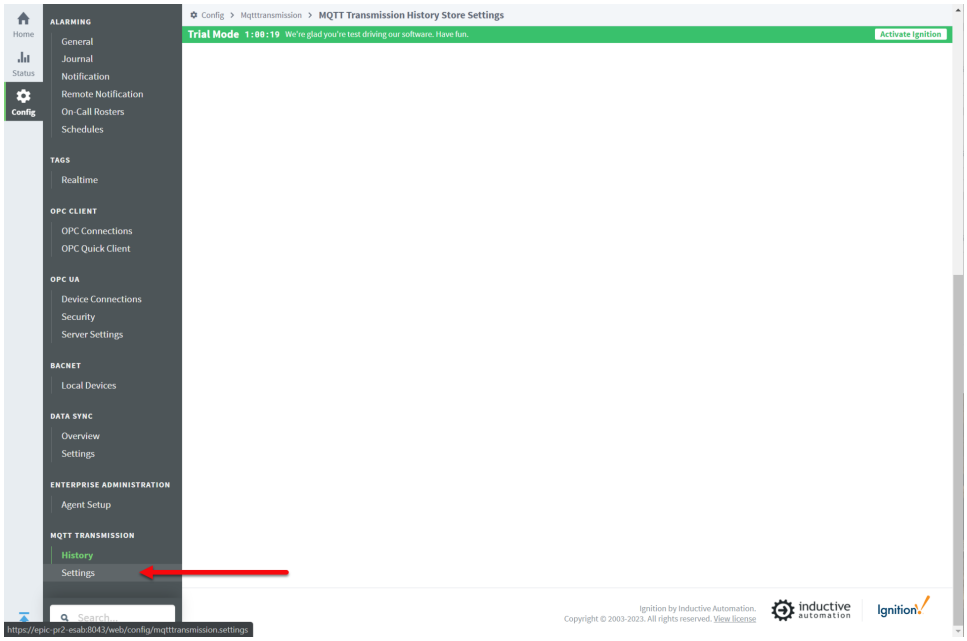


10. Complete the Edge Node Tag Capacity, Device Tag Capacity, Flush Quantity, and Flush Period fields suitable for your application. Good default values for these fields are noted below. Once complete, click Create New

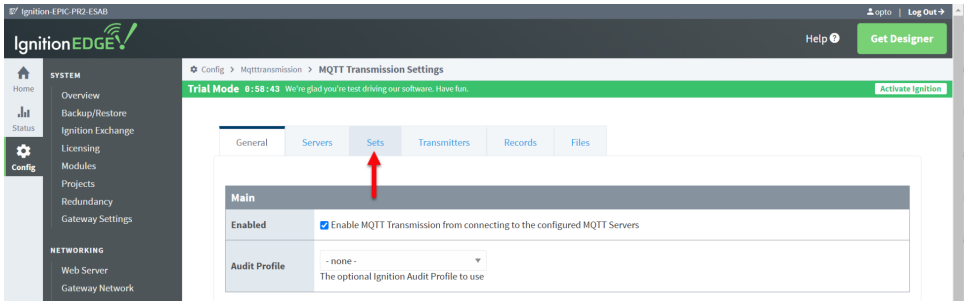
Transmission History Store:



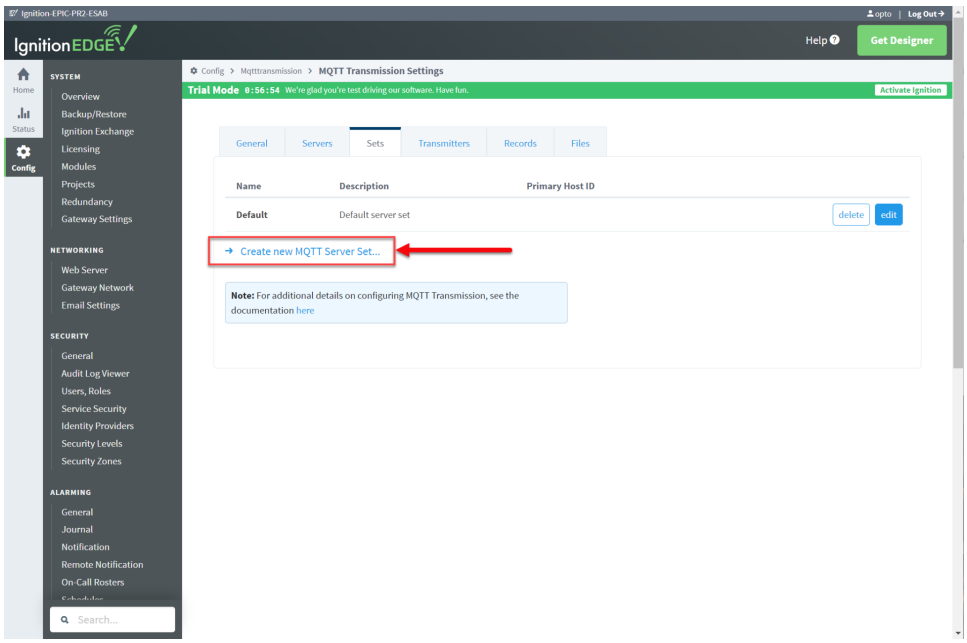
11. In the left nav, scroll down to the MQTT Transmission section, and click on Settings



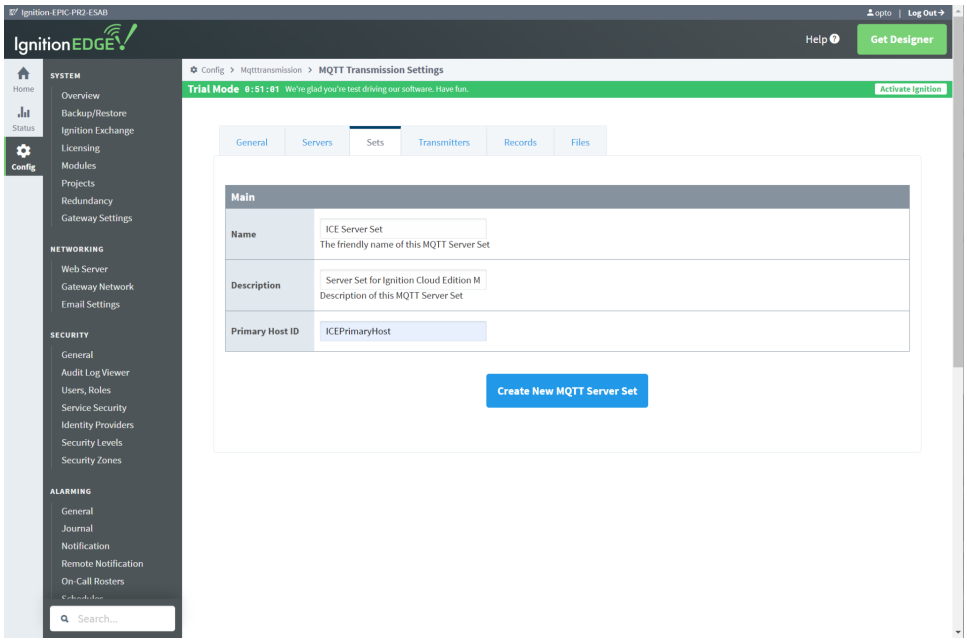
12. From the tabs at the top of the MQTT Transmission Settings page, click on Sets:



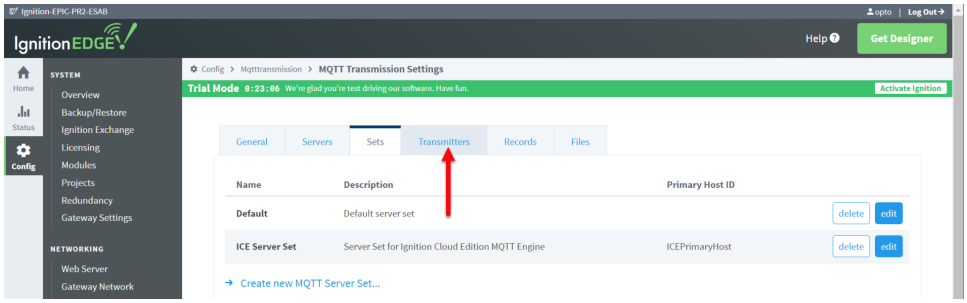
13. Click the blue link that reads Create new MQTT Server Set:



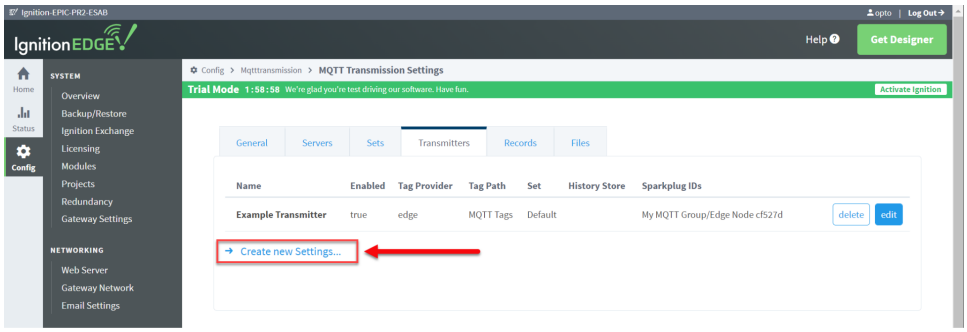
14. The MQTT Transmission Settings - Sets page is where you'll define a Server Set that includes your Primary Host ID, which corresponds to your Ignition server running MQTT Engine.
    a. **Name**: Give your Server Set a name that corresponds to the MQTT server/broker you'll connect to (e.g., ICE Server Set).
    b. **Description**: Add a description if you wish.
    c. **Primary Host ID**: Enter the Primary Host ID of the host application (e.g., ICEPrimaryHost, noted above)
    d. When complete, click the blue Create New MQTT Server Set button.
    ***Note**: In the event the configured host application is unreachable for any reason, store & forward commences, storing data in your previously configured History Store until the host application reports "online" again, at which time stored data is forwarded to the host application.*



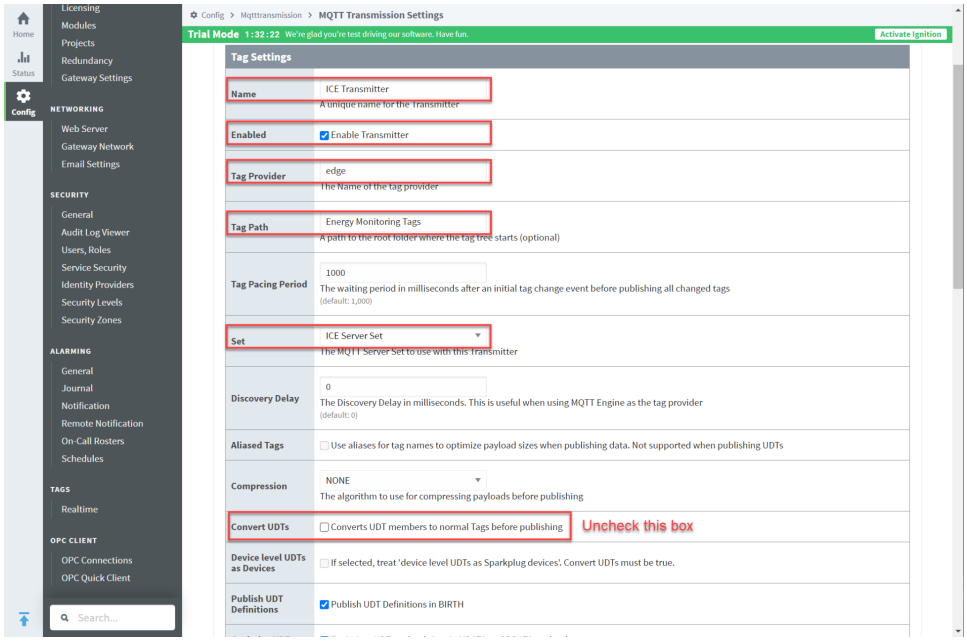
15. From the tabs at the top of the MQTT Transmission Settings page, click on Transmitters.



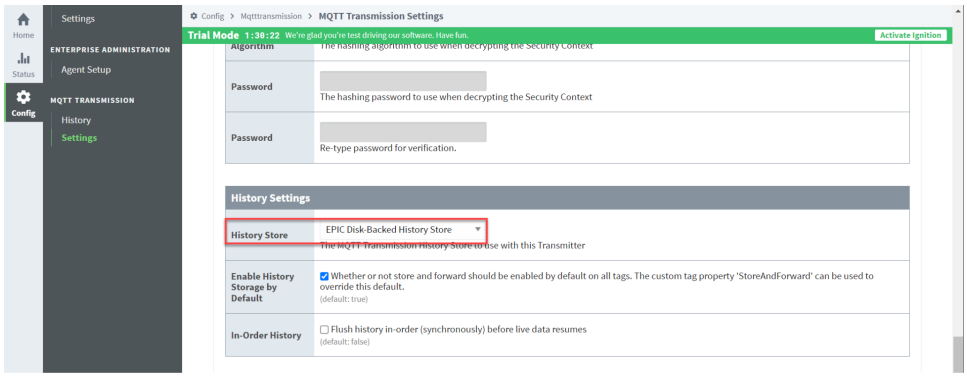
16. Click the blue link that reads Create new MQTT Settings…



17. The MQTT Transmission Setting page is where you define what tags you'd like to publish to the MQTT broker/server. You'll also set the MQTT Namespace (noted above) on this page.
    In the Tag Settings section:
    a. **Name**: Enter a name for this MQTT Transmitter ideally referencing the server (e.g., ICE Transmitter)
    b. **Enabled**: Be sure this is checked
    c. **Tag Provider**: For Ignition Edge, this will always be "edge"
    d. **Tag Path**: This is the folder you created in Designer in previous steps (e.g., Energy Monitoring Tags)
    e. **Set**: Choose the previously configured Server Set (e.g., ICE Server Set)
    f. **Convert UDTs**: UNCHECK this box as we'll be publishing UDTs



    In the History Settings section:
    g. **History Store**: Choose the previously configured History Store (e.g., EPIC Disk-Backed History Store)

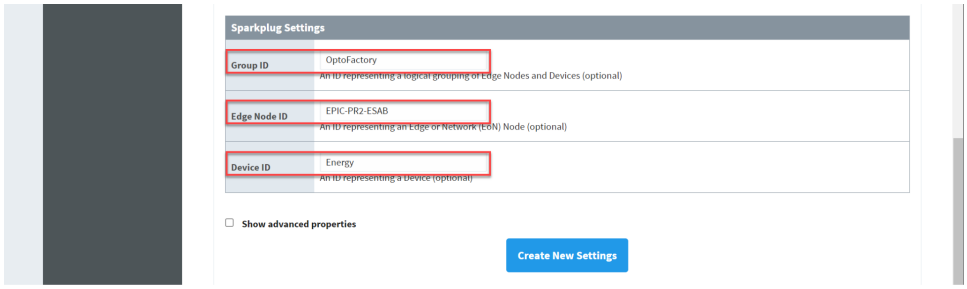


In the Sparkplug Settings section:

h. **Group ID**: Enter your MQTT namespace Group ID; this is the topmost folder the data will publish in under (e.g., OptoFactory)

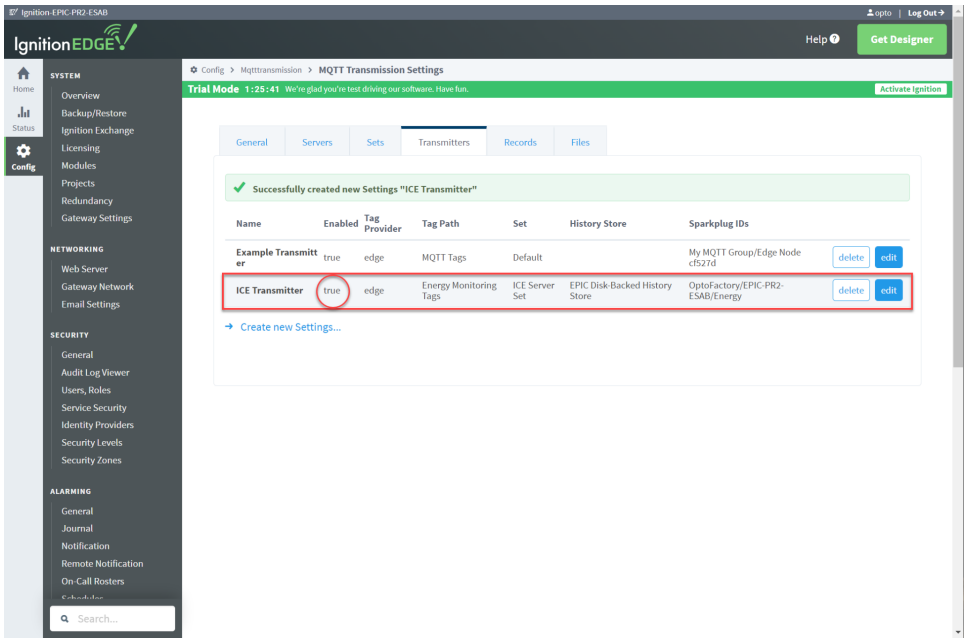
i. **Edge Node ID**: Enter your Edge Node ID; this is a child folder of the above (e.g., EPIC-PR2-ESAB)

j. **Device ID**: Enter the Device ID; this is the 3rd child folder of the above (e.g., Energy)

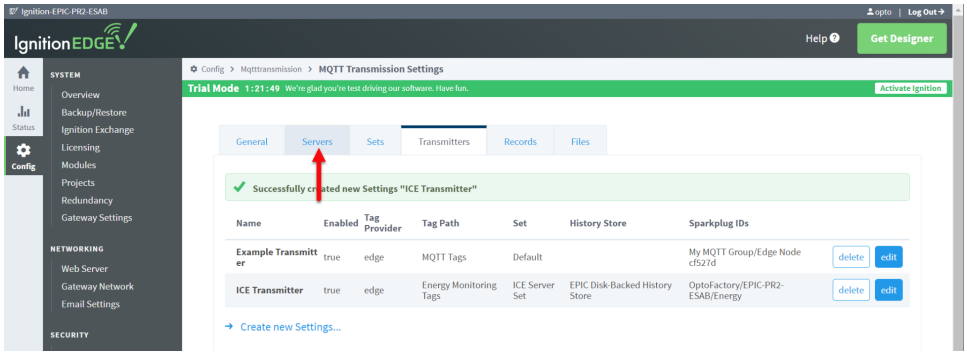


k. Click the blue Create New Settings button

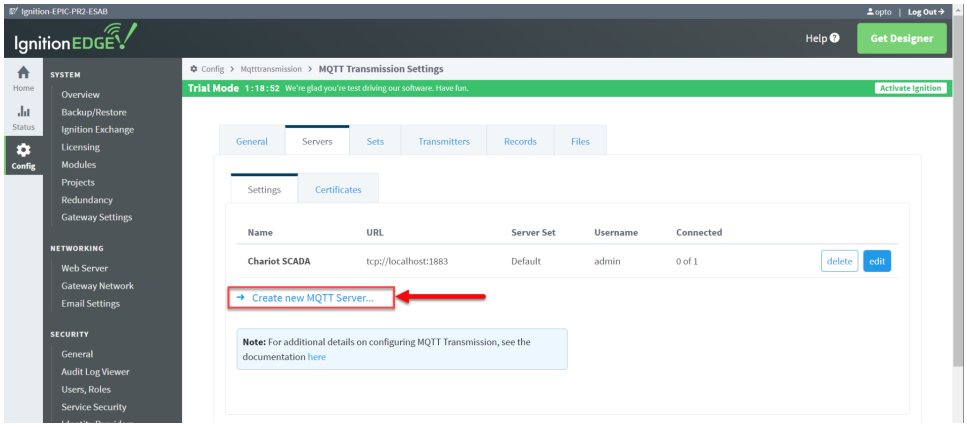
l. The resulting page should show the new Transmitter, and that it is enabled:



18. The final MQTT Transmission Settings step is configuring and connecting to the MQTT broker/server. Click on the tab labeled Servers



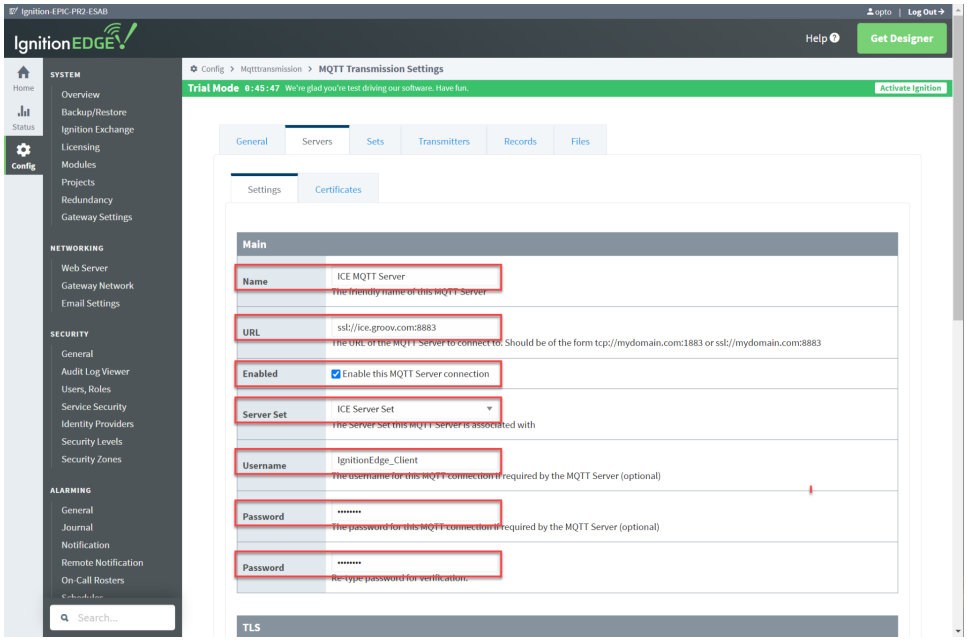
19. Click on the blue link named Create new MQTT Server…



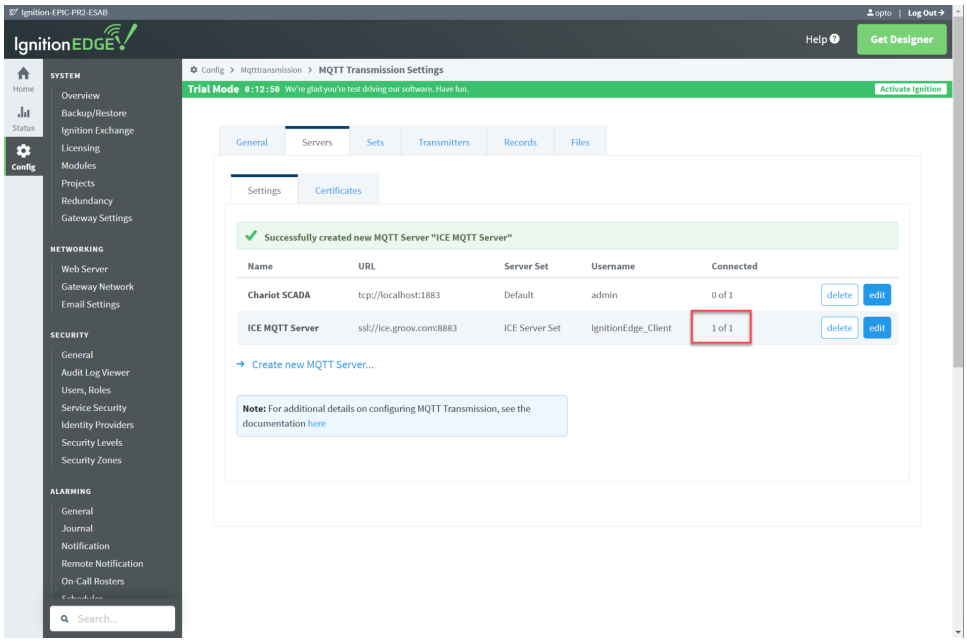
20. Complete the MQTT Transmission Settings - Servers page with the following:
    a. **Name**: The friendly name of your MQTT Server (e.g., ICE MQTT Server)
    b. **URL**: Enter the URL to your MQTT server; for secure connections, prepend the URL with ssl:// and append the URL with port 8883 (e.g., ssl://ice.groov.com:8883)
    c. **Enabled**: Ensure this box is checked
    d. **Server Set**: Choose the previously configured Server Set (e.g., ICE Server Set)
    e. **Username**: Enter the Username credential for your MQTT server (e.g., IgnitionEdge_Client)
    f. **Password:** Enter the Password credential for your MQTT server, twice (e.g., opto22!!)
    g. Scroll to the bottom of the page, and click the Create New MQTT Server button

***Note**: If your Ignition Cloud Edition or MQTT broker's Server Certificate is not signed by a Global CA, you may need to install a valid Client Certificate for your instance on this page. While not recommended, you may*

elect to transmit your MQTT messages over an unsecured network connection for testing.



21. The resulting page should show the status of your MQTT connection. You're looking for "1 of 1" in the Connected column

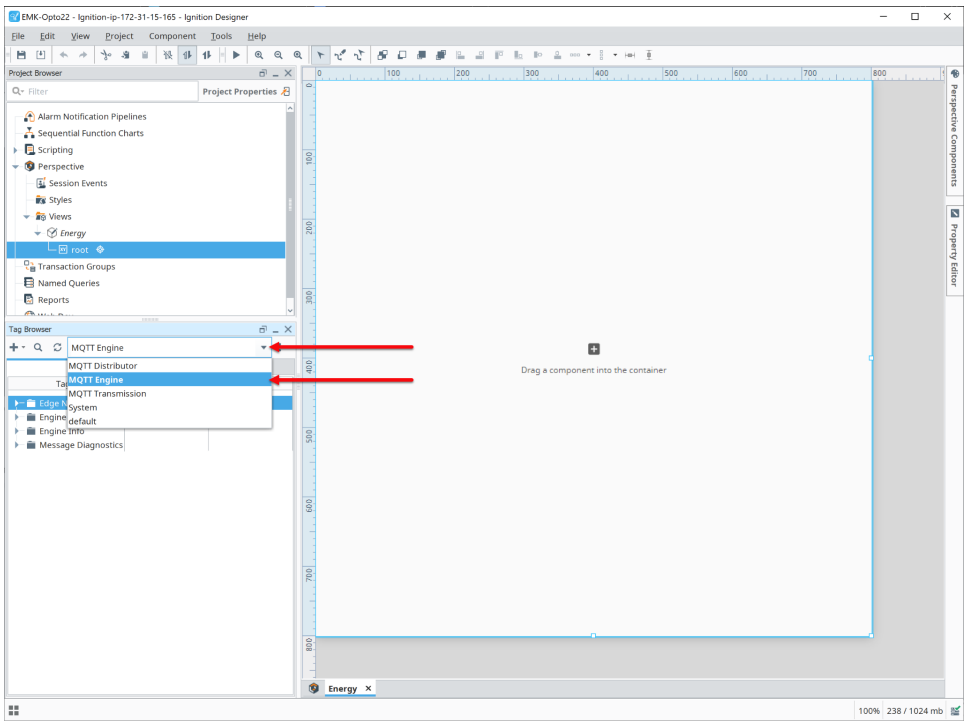


Confirming UDTs Publishing into Ignition Cloud Edition

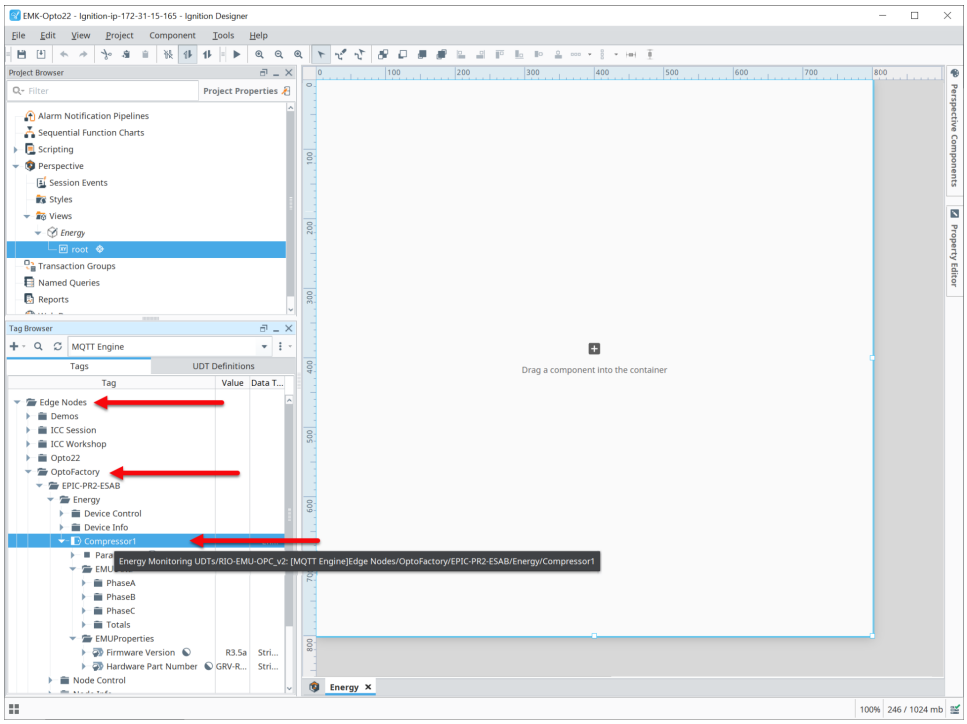
While not part of this exercise, it's helpful to check your primary host application to confirm your Energy Monitoring UDTs are publishing successfully. In this example, I'm using Ignition Cloud Edition. So, I'll log into my Ignition Cloud Edition Designer, and view the tags coming in

1. Log into your Designer for Ignition Cloud Edition.

2. In the Tag Browser pane, choose the Tag Provider named MQTT Engine



3. Expand the folder named Edge Nodes to see all the Group IDs publishing. Click on the Group ID we're publishing on (e.g., OptoFactory) and expand each folder to reveal the real-time data coming into MQTT Engine



4. At this point, you can start building a Vision or Perspective dashboard for your energy data. You can also set up history, alarms, reporting functions, and much, much more–leveraging all the power of Ignition for your energy and power monitoring needs.